

EXPLAINABLE ARTIFICIAL INTELLIGENCE

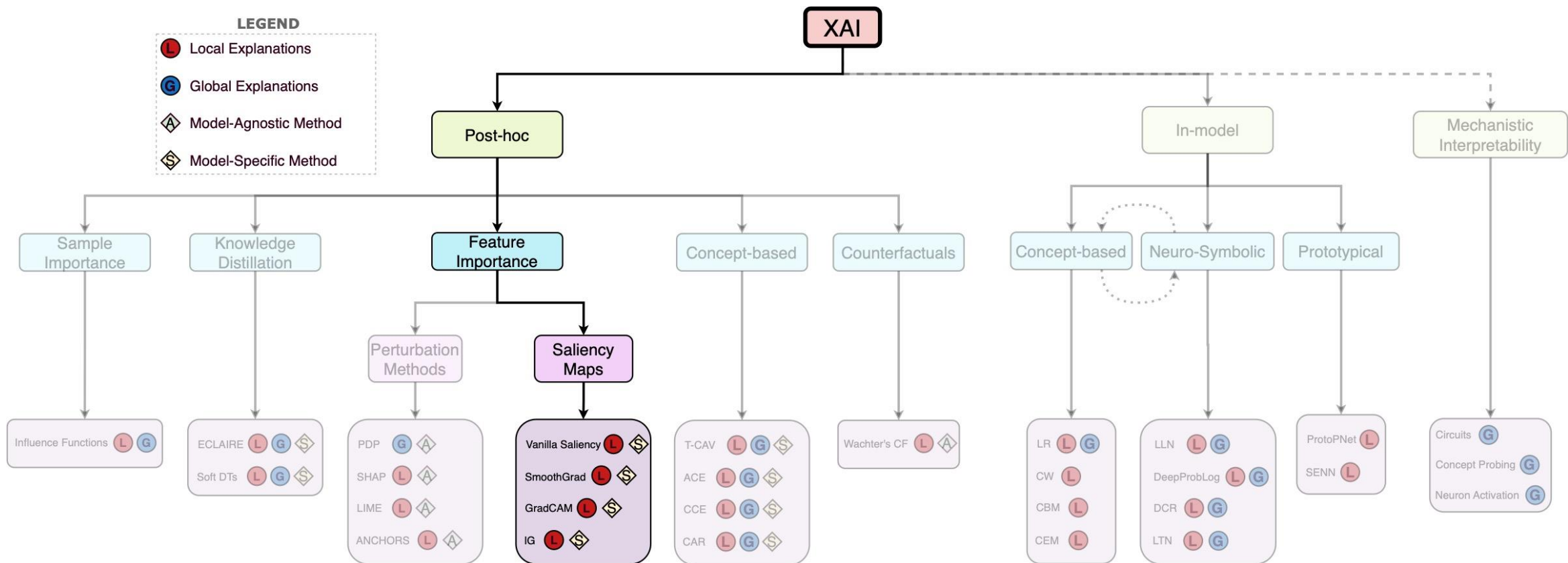
L193 – Lecture 3 – Lent 2025



UNIVERSITY OF
CAMBRIDGE

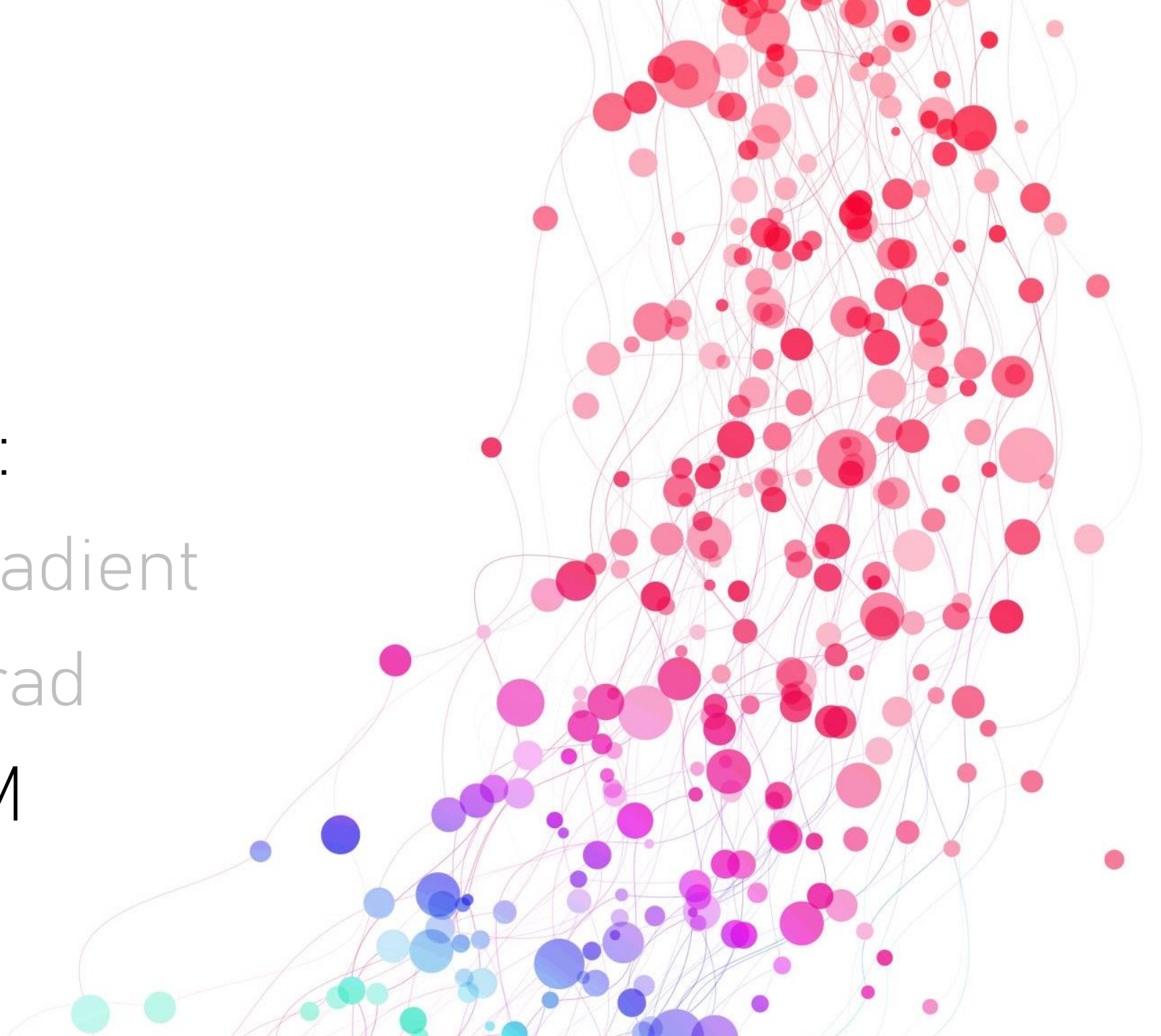


WHERE WE LEFT THINGS



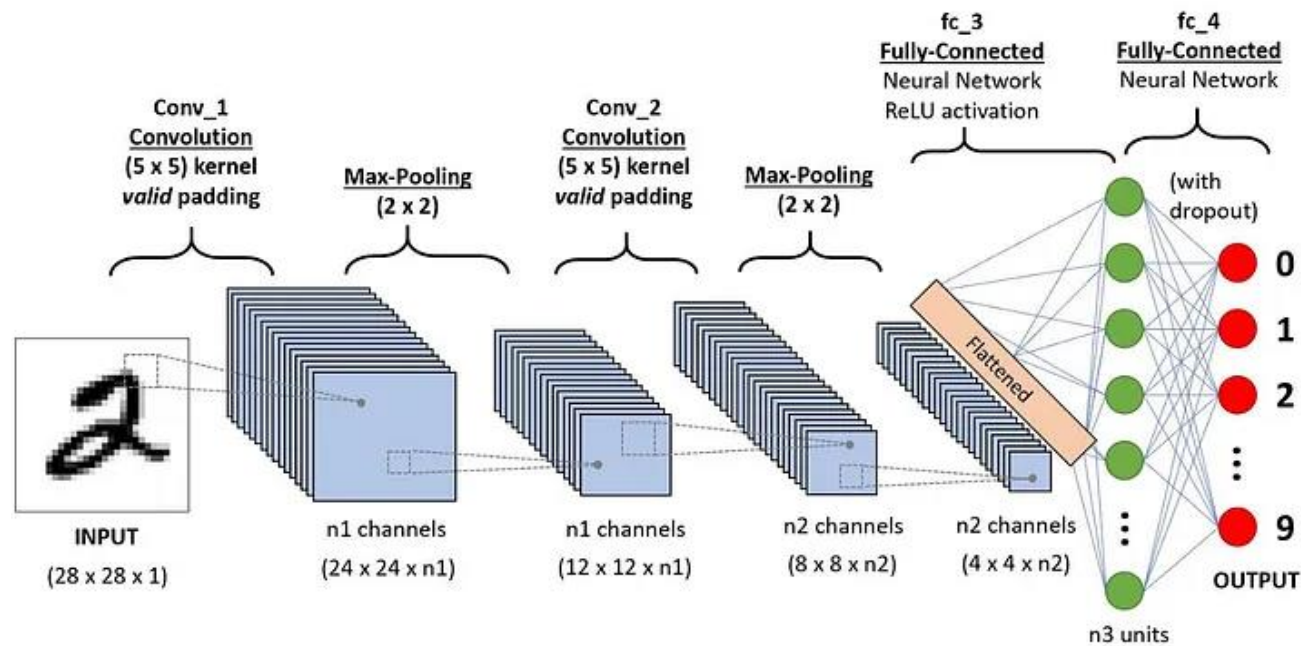
Saliency Maps:

- Vanilla Gradient
- SmoothGrad
- Grad-CAM



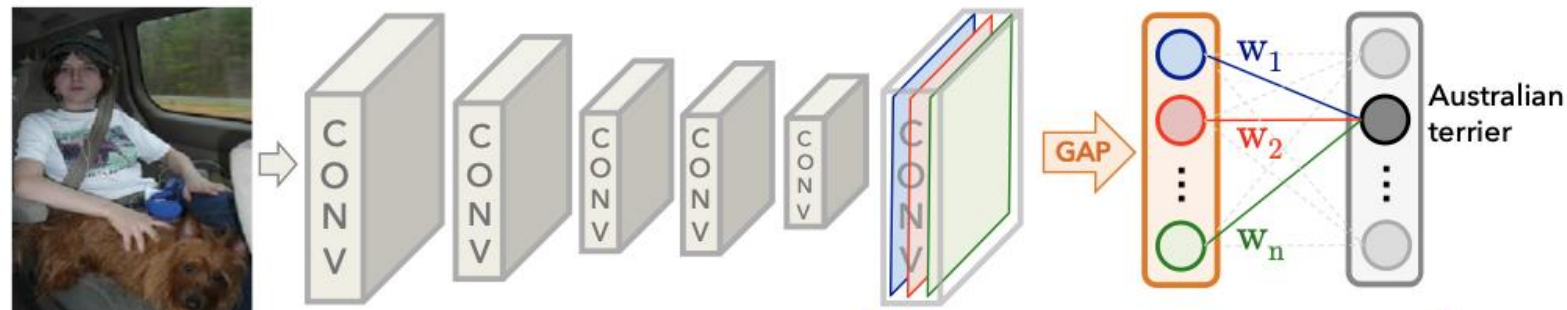
CONVENTIONAL CNNs

The **output feature maps** from the last convolutional layer are typically **flattened** and then passed into one or more **fully connected layers** before reaching the **final classification layer**



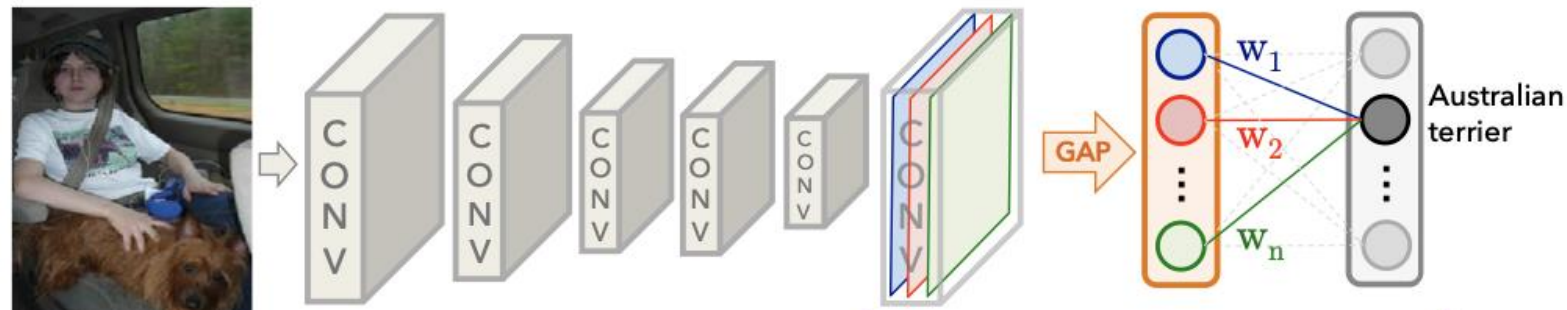
CNN WITH GLOBAL AVERAGE POOLING (GAP)

In architectures using *Global Average Pooling* (GAP) [1], the feature maps are summarized per channel using GAP, and the resulting feature vector is directly passed to the final classification layer



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$$\text{GAP}^{(k)} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W A_{ij}^{(k)}$$

Height and width of feature map

Activation value at pixel (i, j) for the k^{th} feature map

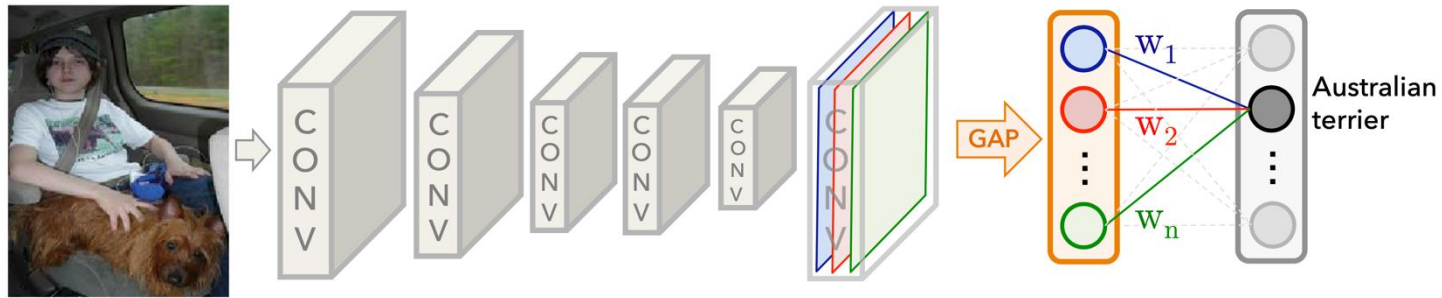
$$y^c = \sum_{k=1}^K w_k^c \cdot \text{GAP}^{(k)} + b^c$$

linear weight associated with feature map k for class c

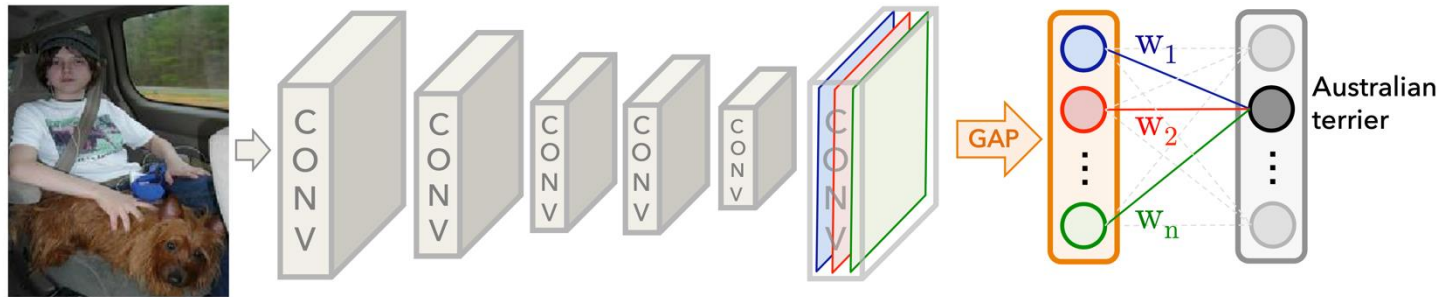
CLASS ACTIVATION MAPPING (CAM) I

- Advantages of using GAP:
 - It is parameter-free and reduces dimensionality → helps **avoid overfitting**
 - It sums out the spatial information → robust to spatial transformation (**spatial invariance**)
 - Most importantly is its **interpretability** properties: the weight of the linear layer directly translate to importance of spatial features for a class → the very idea in CAM [1]

CLASS ACTIVATION MAPPING (CAM) II

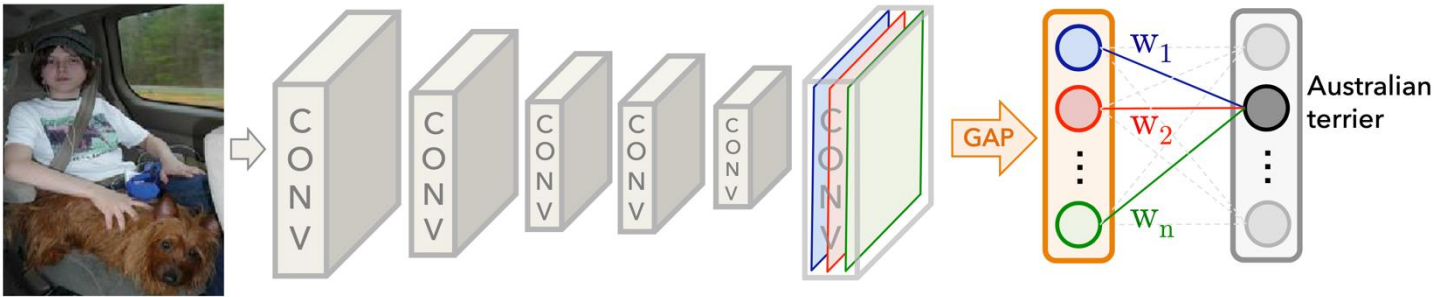


CLASS ACTIVATION MAPPING (CAM) II



Assumption: I have normalised the feature maps and have converted them each to a heatmap

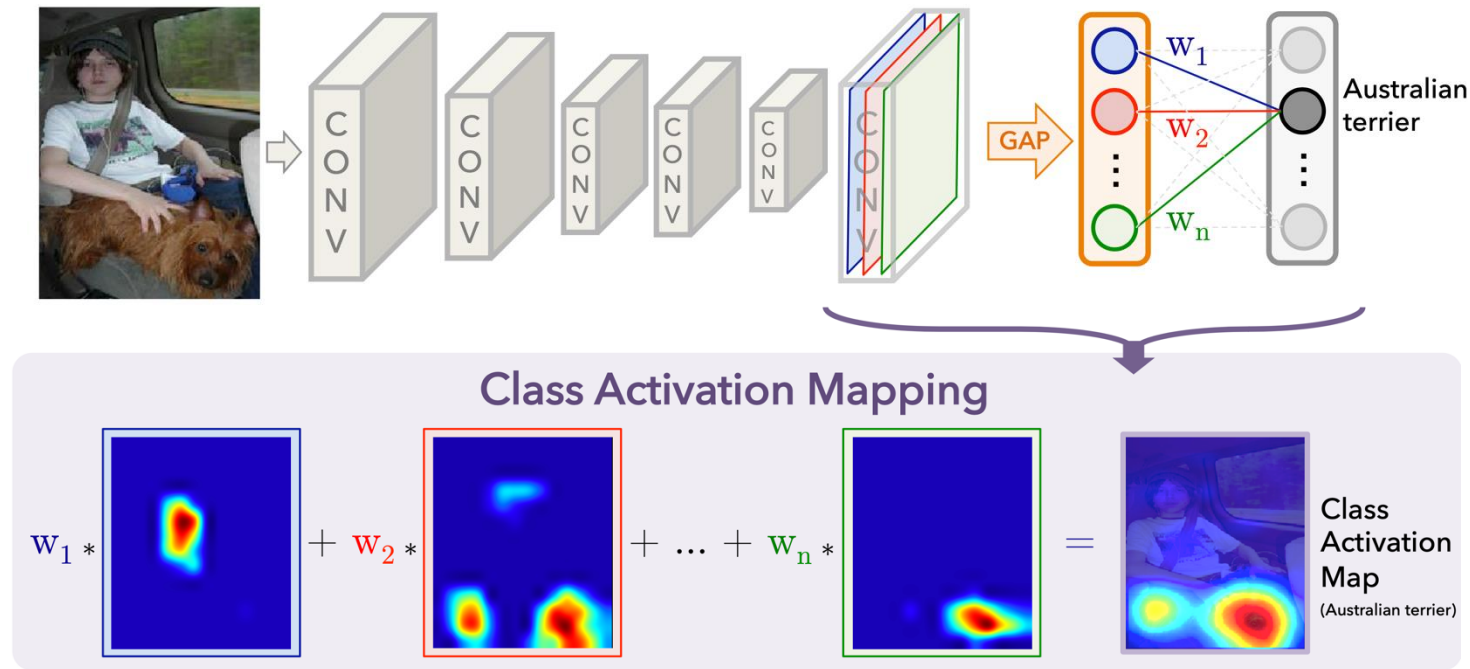
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Question: Can I overlay the activation maps on the original image?

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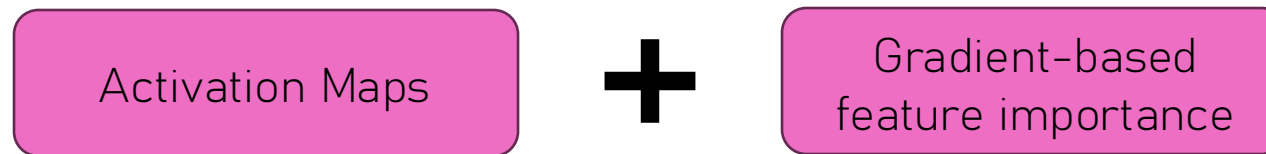
$$\text{CAM heatmap for class } c : L_{CAM}^c = \sum_{k=1}^K w_k^c A^{(k)}$$

GRAD-CAM

- CAM requires GAP followed by a linear layer → very architecture specific
- Grad-CAM [1] is a generalisation of CAM that works with **any** architecture!

GRAD-CAM

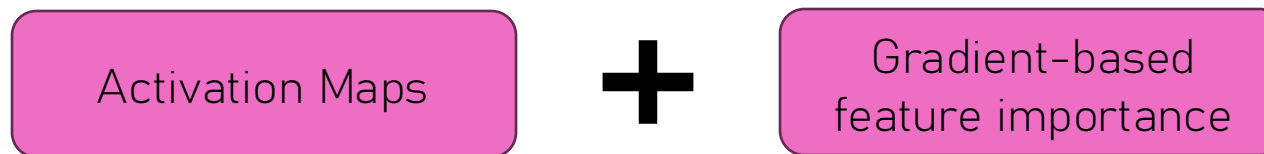
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[1] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D. and Batra, D., 2017. Grad-cam: Visual explanations from deep networks via gradient-based localization. In Proceedings of the IEEE international conference on computer vision (pp. 618-626).

GRAD-CAM

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Unlike other gradient methods, the gradient is not back-propagated all the way to the input layer, but to the **last convolutional layer** of the CNN (feature maps)

[1] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D. and Batra, D., 2017. Grad-cam: Visual explanations from deep networks via gradient-based localization. In Proceedings of the IEEE international conference on computer vision (pp. 618-626).

GRAD-CAM FORMULATION

- Step 1: compute gradient of class score w.r.t. feature maps

$$\alpha_k^c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \frac{\partial y^c}{\partial A_{ij}^{(k)}} \quad \text{Gradient of class score w.r.t. pixel } (i,j) \text{ in feature map } A^k$$

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$$L_{Grad-CAM}^c = ReLU\left(\sum_{k=1}^K \alpha_k^c A^{(k)}\right)$$

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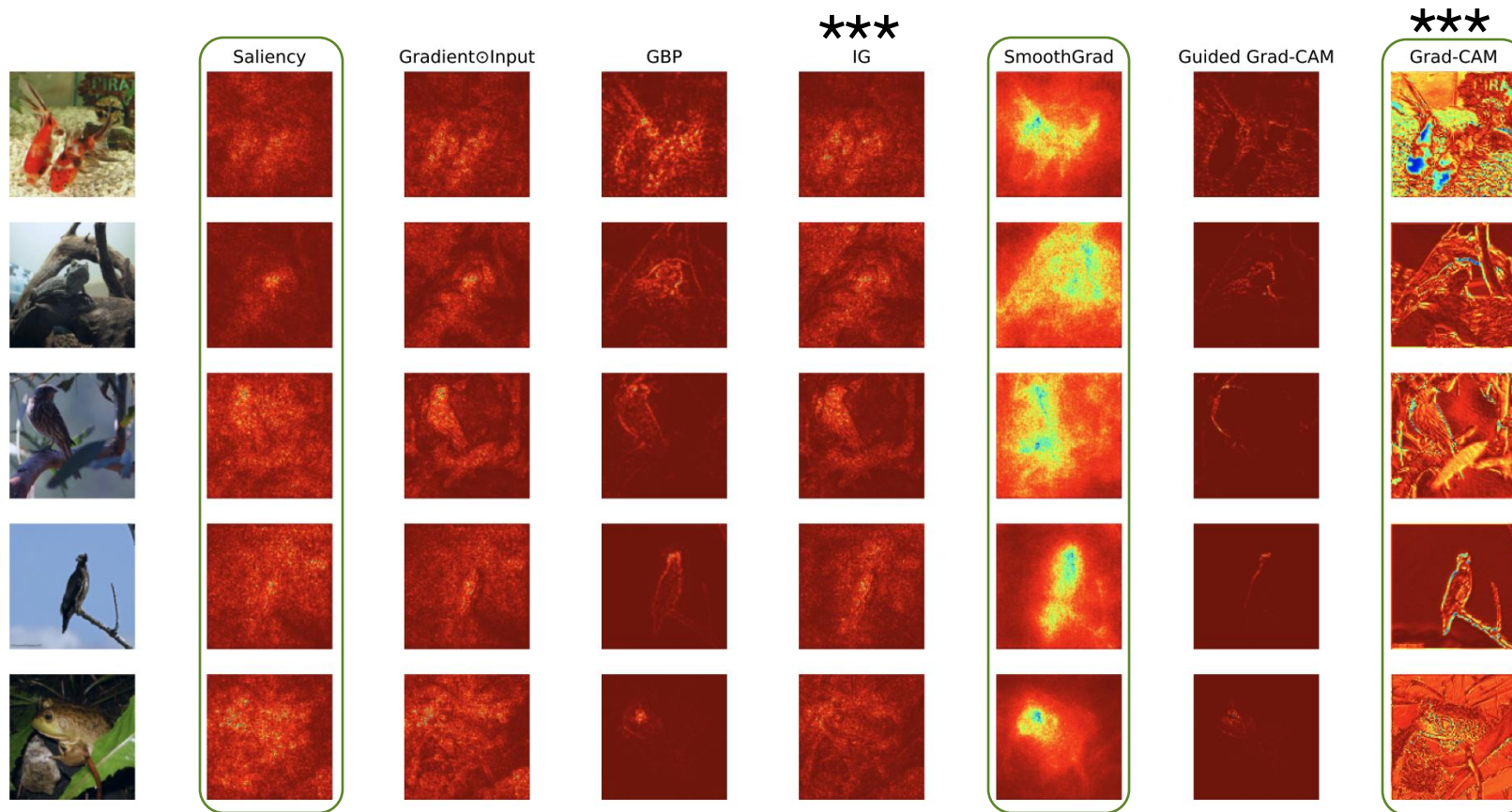
- What's the role of ReLU here?

CAM

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FAMILY OF GRADIENT-BASED METHODS



Useful Python Library



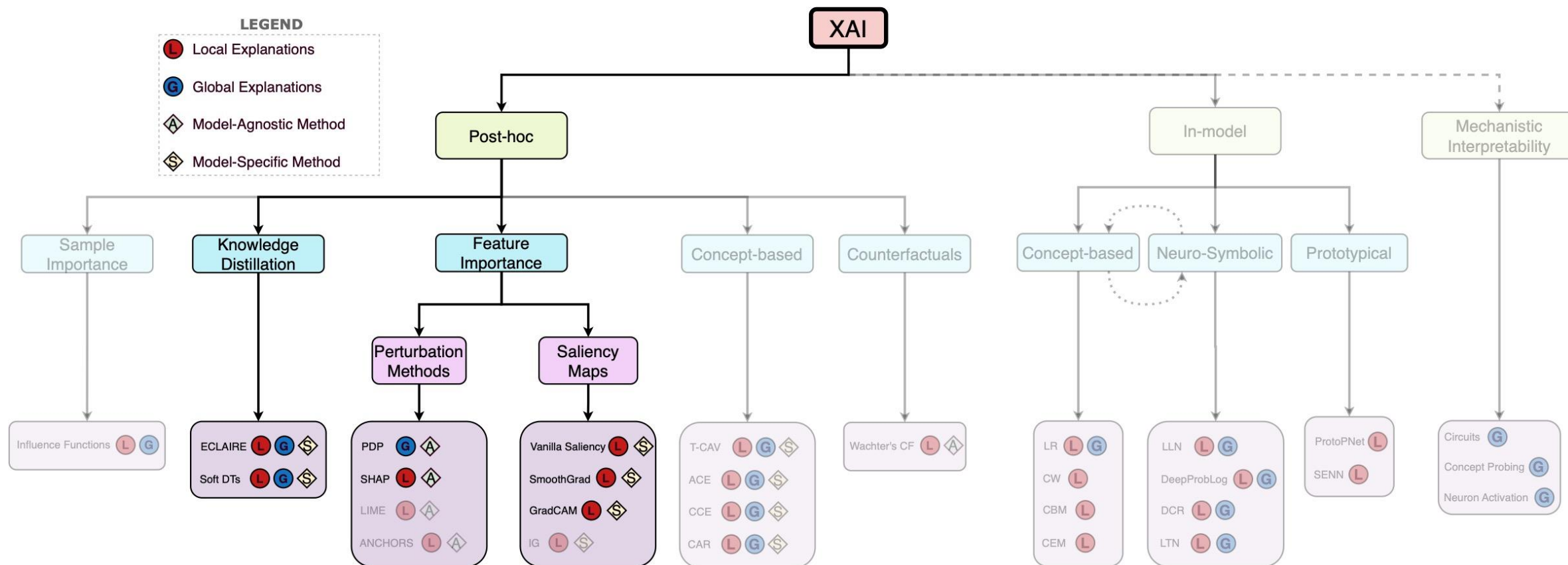
<https://github.com/albermax/innvestigate>

Comparative review of gradient-based methods [1]

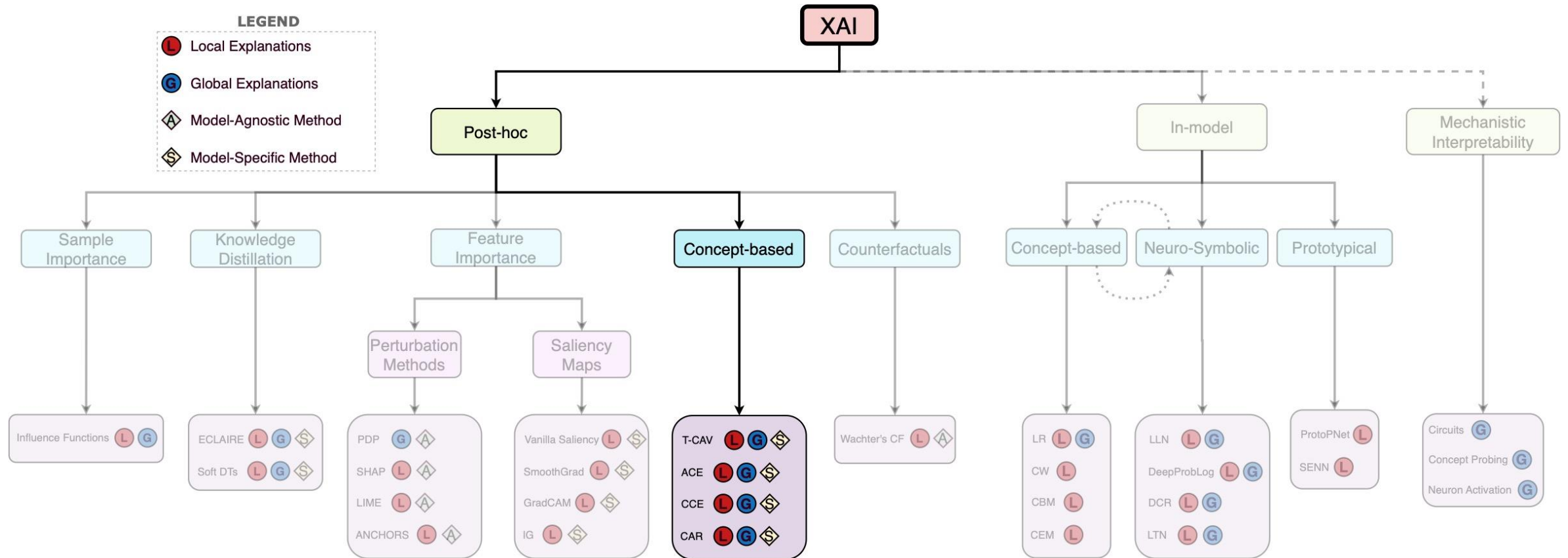
[1] Nielsen, Ian E., et al. "Robust explainability: A tutorial on gradient-based attribution methods for deep neural networks." *IEEE Signal Processing Magazine* 39.4 (2022): 73-84.

Image taken from [1].

THE STORY SO FAR



DIFFERENT TYPES OF EXPLANATIONS



WHAT'S WRONG WITH FEATURE ATTRIBUTION?

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1. Low-level features like individual pixels are **not always semantically meaningful**:



Can you guess what this is?

WHAT'S WRONG WITH FEATURE ATTRIBUTION?

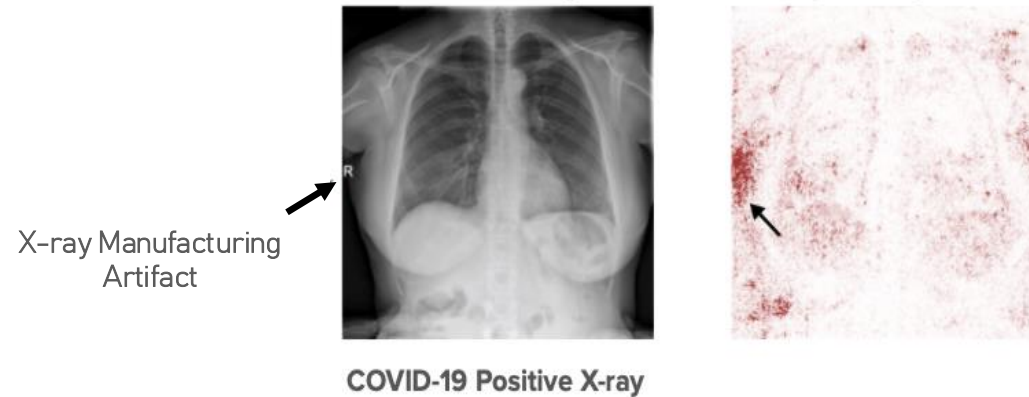
1. Low-level features like individual pixels are **not always semantically meaningful**:



Limes!

WHAT'S WRONG WITH FEATURE ATTRIBUTION?

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WHAT'S WRONG WITH FEATURE ATTRIBUTION?

2. Feature maps lack of **actionability!**



What does this really tell you about **how** the model made a prediction?

WHAT'S WRONG WITH FEATURE ATTRIBUTION?

3. They are susceptible to **adversarial attacks** [1,2]



More on this next week!

[1] Dombrowski, Ann-Kathrin, et al. "Explanations can be manipulated and geometry is to blame." *Advances in Neural Information Processing Systems* 32 (2019).

[2] Ghorbani, Amirata, Abubakar Abid, and James Zou. "Interpretation of neural networks is fragile." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 33. No. 01. 2019.

WHAT'S WRONG WITH FEATURE ATTRIBUTION?

How can we go around the limitations of feature attribution?

- Other forms of explainability (e.g., knowledge distillation)
- **Concept-based explainability!**

WHAT ARE CONCEPTS?

Concepts are **high-level** and semantically **meaningful** units of information

Task:
bird species



Explanation of the prediction:

- wing colour
- beak length
- tail shape

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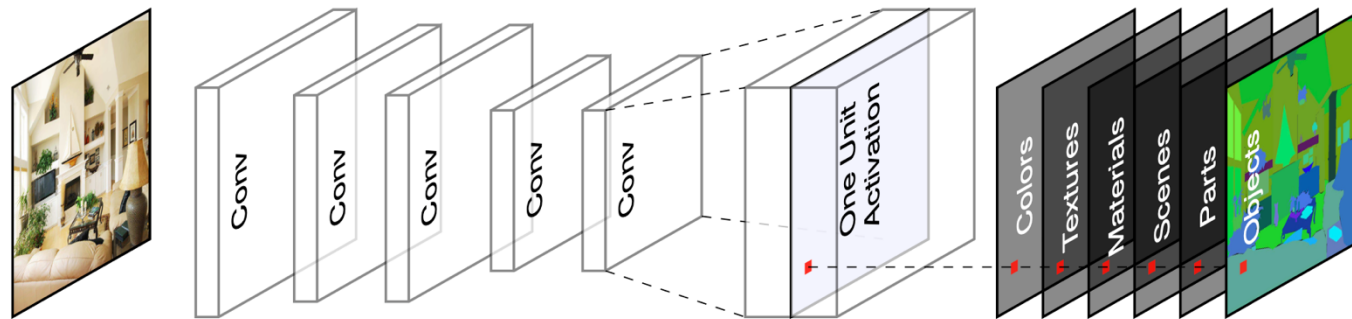
Explanation of the prediction:

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Concepts are useful when they are **used by domain experts** to communicate or explain things to one another

DO NEURAL NETWORKS NATURALLY LEARN CONCEPTS?

Evidence generally suggests that is the case: lower levels are detecting texture or surface, whilst higher levels learn more semantically meaningful concepts [1,2]



Next week you will hear in more detail about this :)

[1] Bau, David, et al. "Network dissection: Quantifying interpretability of deep visual representations." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017

[2] Fong, Ruth, and Andrea Vedaldi. "Net2vec: Quantifying and explaining how concepts are encoded by filters in deep neural networks." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018.

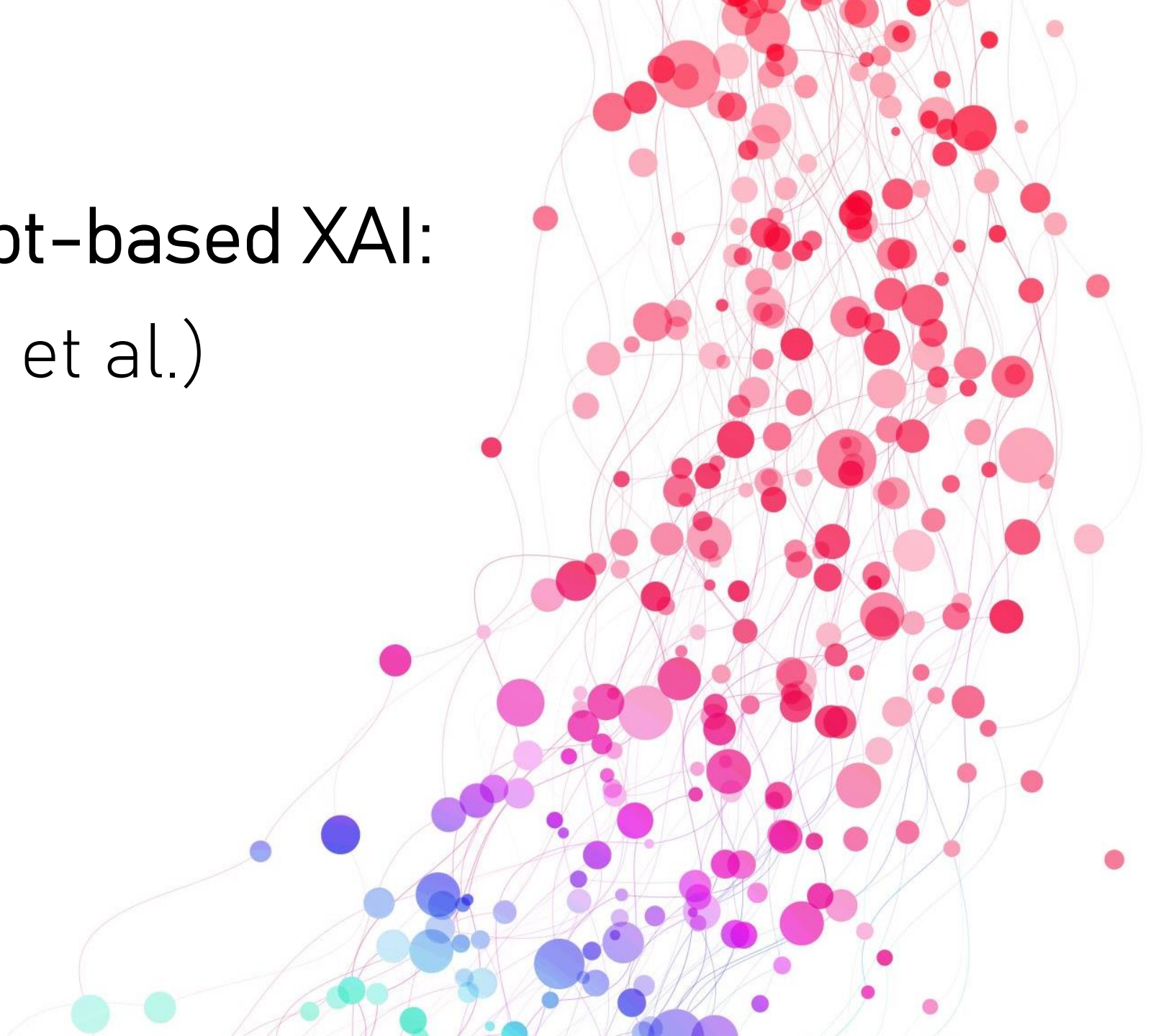
Image taken from [1]

POST-HOC CONCEPT-BASED EXPLAINABILITY



Post-hoc Concept-based XAI:

- T-CAV (Kim et al.)



T-CAV (CONCEPT ACTIVATION VECTOR): INTUITION

- **T-CAV** [1] provides **global** explanations for a class of interest



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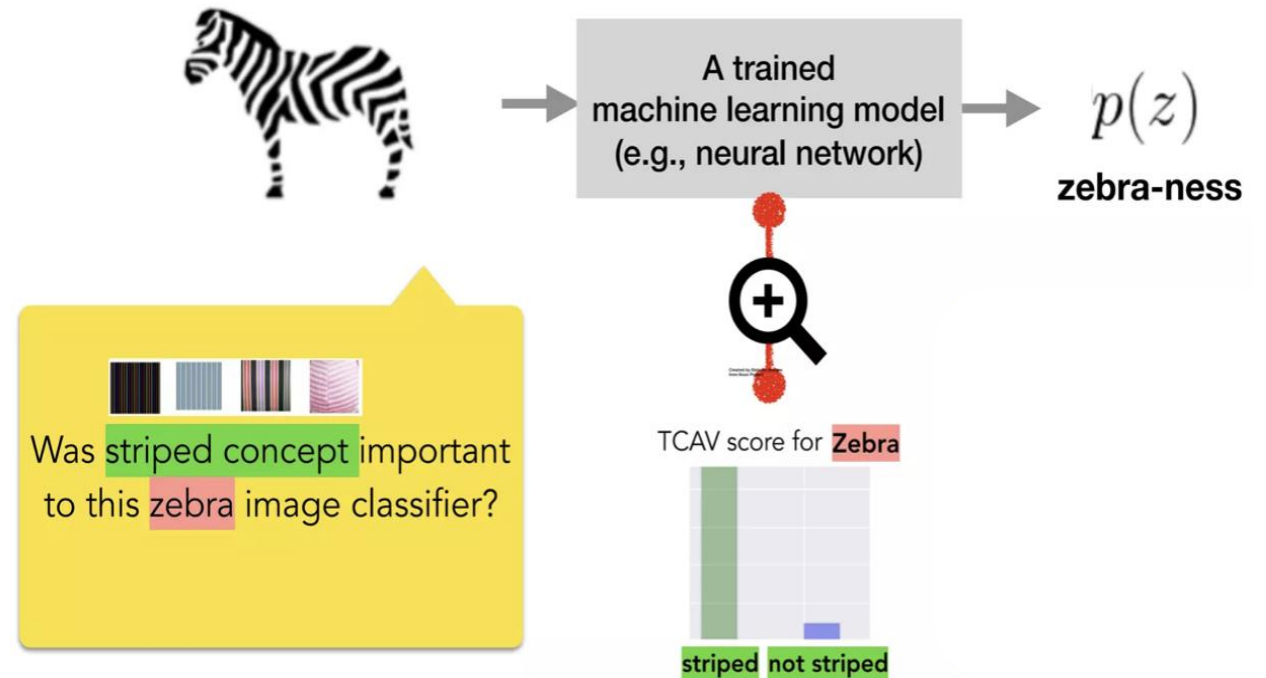
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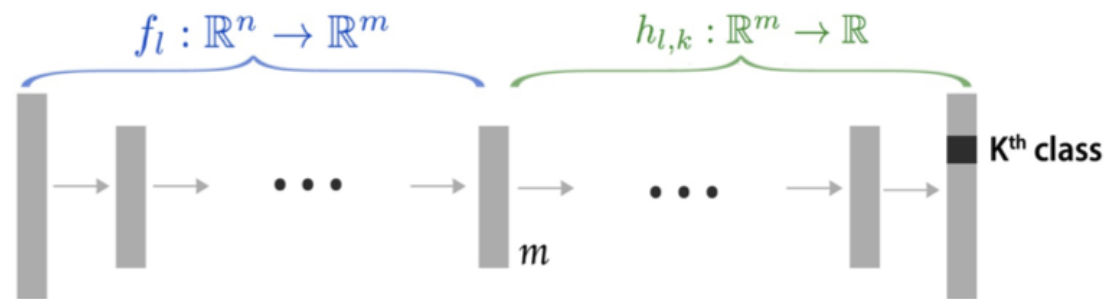
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How sensitive prediction of zebra
is to the presence of stripes



T-CAV: FORMULATION

Step 1: Choose an intermediate layer $f_l: \mathbb{R}^n \rightarrow \mathbb{R}^m$ with m neurons

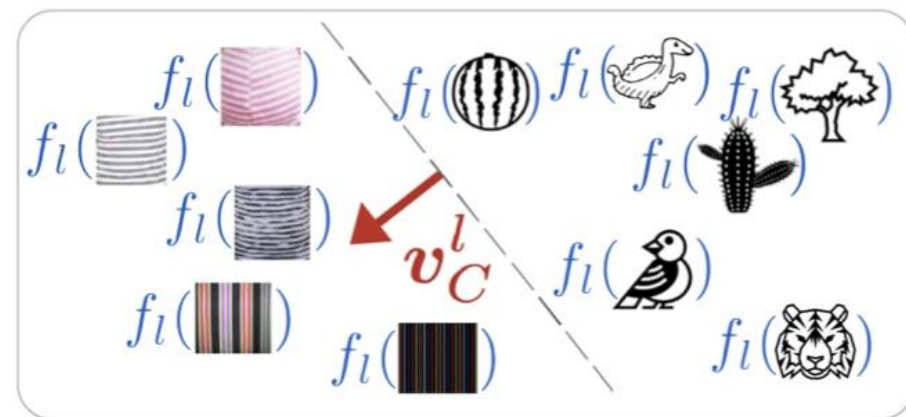


T-CAV: FORMULATION

Step 1: Choose an intermediate layer $f_l: R^n \rightarrow R^m$ with m neurons

Step 2: Learn the *Concept Activations Vectors* (CAVs)

- Train a linear classifier to distinguish between the activations of concept's examples and random ones
- The CAV is the **vector orthogonal** to the classification boundary v_C^l



T-CAV: T-CAV SCORES

Step 3: Getting **importance scores** from CAVs

- T-CAV gauges the *sensitivity* of class k to concept $\mathcal{C}$
- Given a sample's latent representation and the CAV, **how do you think we should gauge this sensitivity?**

T-CAV: T-CAV SCORES

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When in doubt, 72% of the time, the answer will be “**derivatives**”

T-CAV: T-CAV SCORES

Step 3: Getting **importance scores** from CAVs

- TCAV uses the **directional derivative**, $S_{C,k,l}(x)$ to gauge how much a classification changes with a change in a concept

Intermediate representation of
 at layer l

$$S_{C,k,l}(x) = \nabla h_{l,k}(f_l(x)) \cdot v_C^l = \nabla h_{l,k}(f_l(\text{zebra})) \cdot v_C^l$$

Output function CAV for concept C (e.g., stripes)

The rate of change of output function as we move in the direction of a concept from data point 

T-CAV: T-CAV SCORES

Step 3: Getting **importance scores** from CAVs

- TCAV uses the **directional derivative**, $S_{C,k,l}(x)$ to gauge how much a classification changes with a change in a concept

T-CAV score is the fraction of k -class inputs that are positively influenced by concept C :

$$\text{TCAV}_{Q_{C,k,l}} = \frac{|\{x \in X_k : S_{C,k,l}(x) > 0\}|}{|X_k|}$$

X_k : inputs
with label k

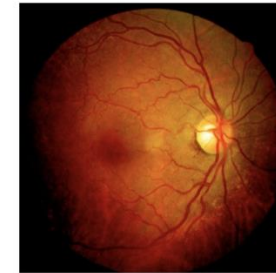


Question: why not also consider the negative influences?

T-CAV: APPLICATION

Medical diagnosis – image data:

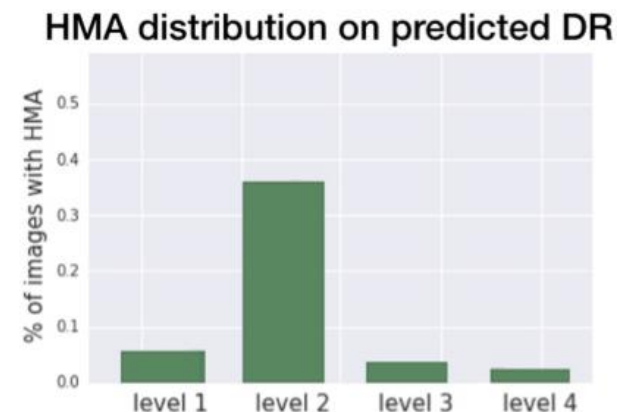
- **Task:** predicting diabetic retinopathy (DR) – level 0 (no DR) to 4 (proliferative)
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- Different concepts are more prominent at different DR levels



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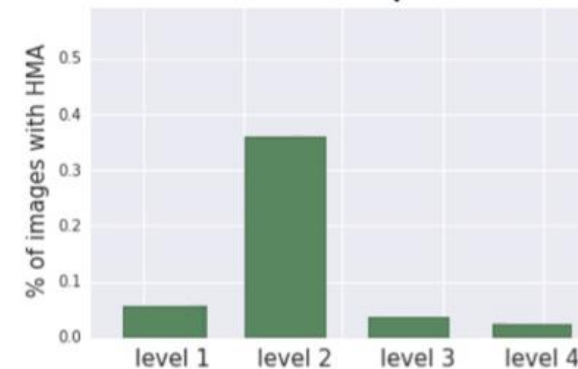


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HMA distribution on predicted DR



DR level 1 Retina



TCAV for DR level 1

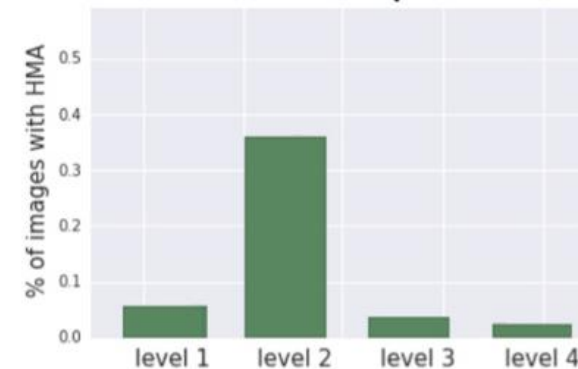


T-CAV: APPLICATION

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DR level 1 Retina



TCAV for DR level 1



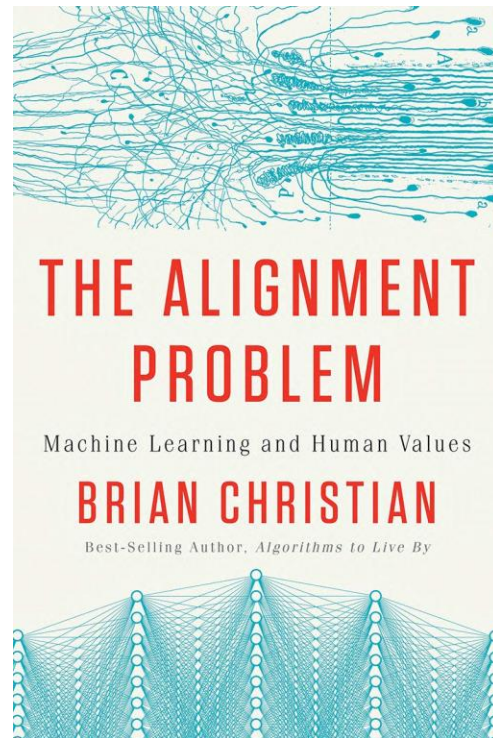
Model error exposed: TCAV score for HMA feature is too high for level 1

THE COST OF BEING GREAT

T-CAV is great!

THE COST OF BEING GREAT

T-CAV is great! So great it is even discussed in a science communication book



THE COST OF BEING GREAT

However, T-CAV requires large sets of examples of each concept of interest:



For example, when finding the influence of the concept “stripes” for a DNN, T-CAV requires a set of samples that all have the concept “stripes”

This is what we call a **concept-supervised approach**!

THE COST OF BEING GREAT

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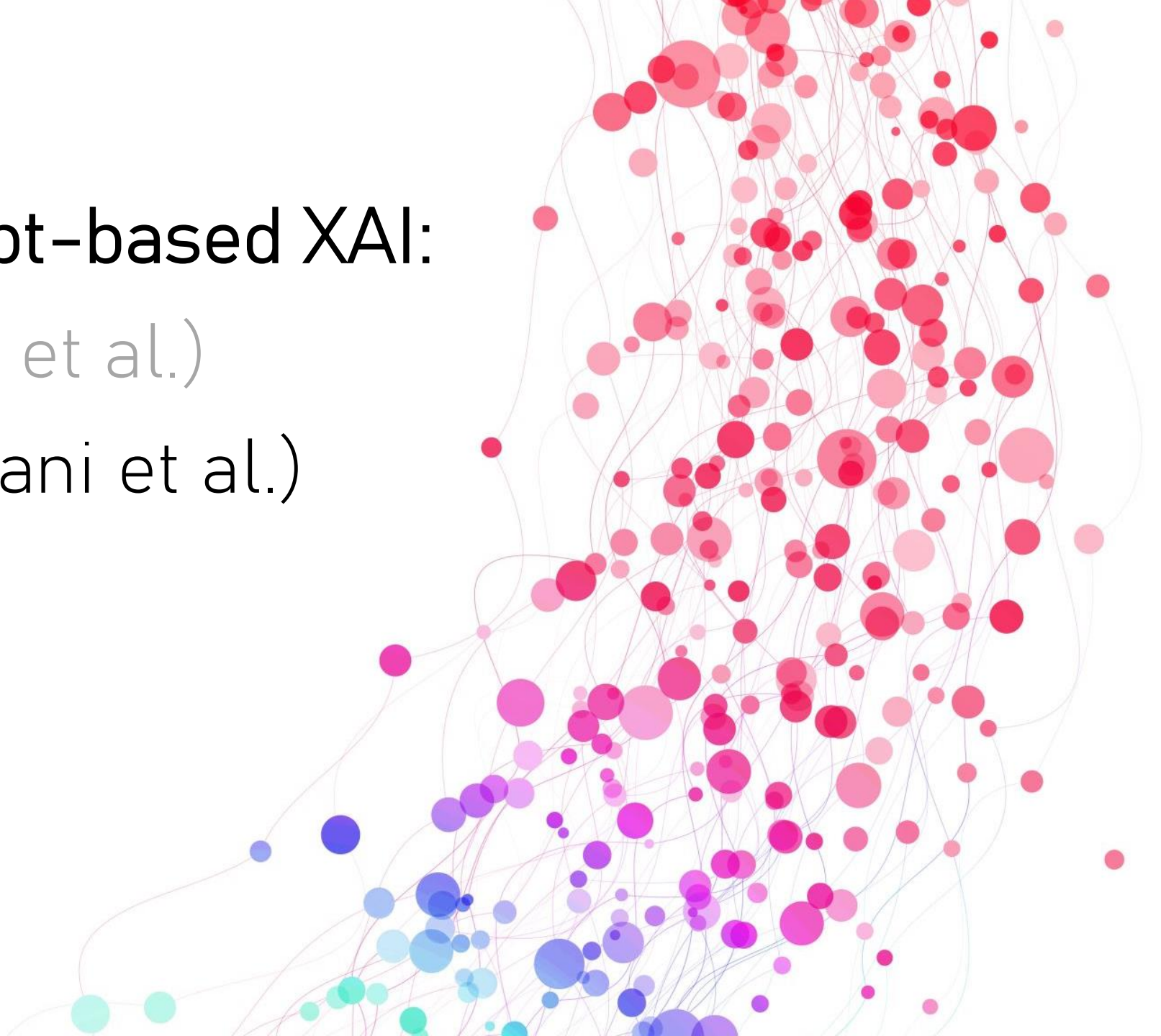


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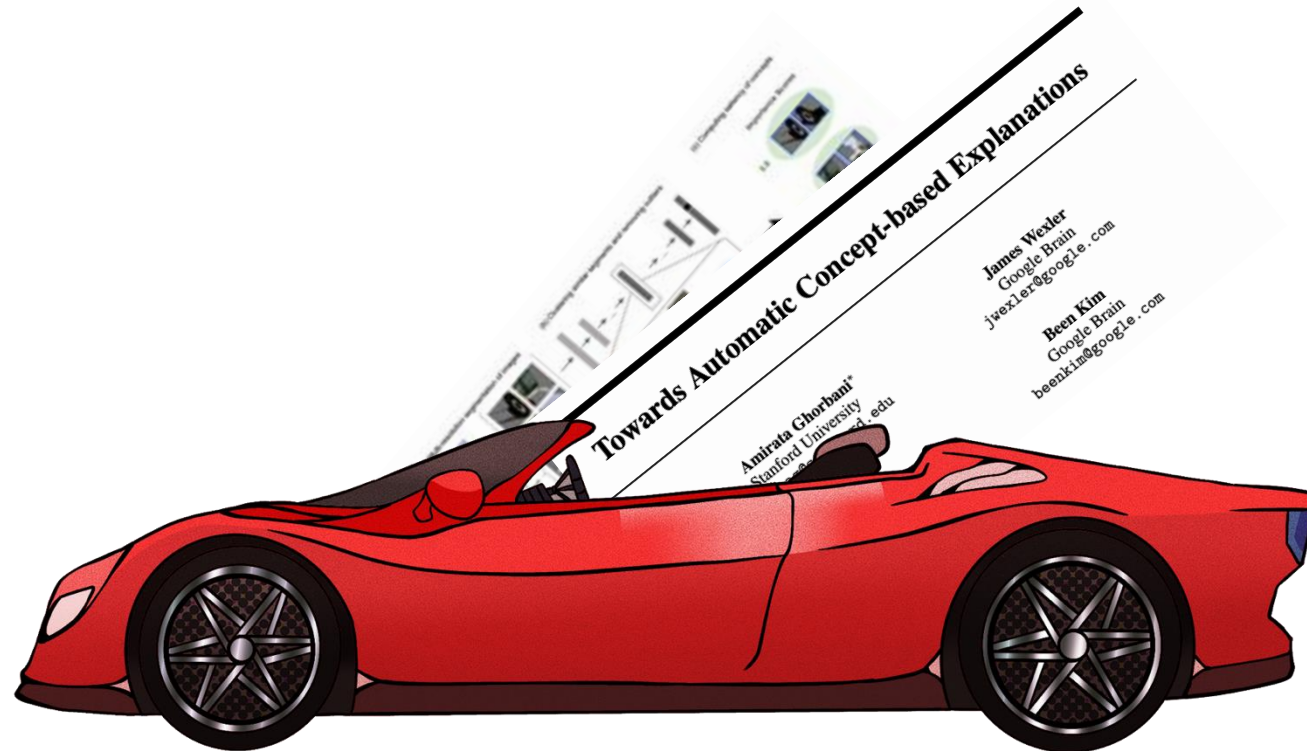
But, obtaining concept labels can be **expensive** and **intractable**

Post-hoc Concept-based XAI:

- T-CAV (Kim et al.)
- ACE (Ghorbani et al.)



GOING UNSUPERVISED

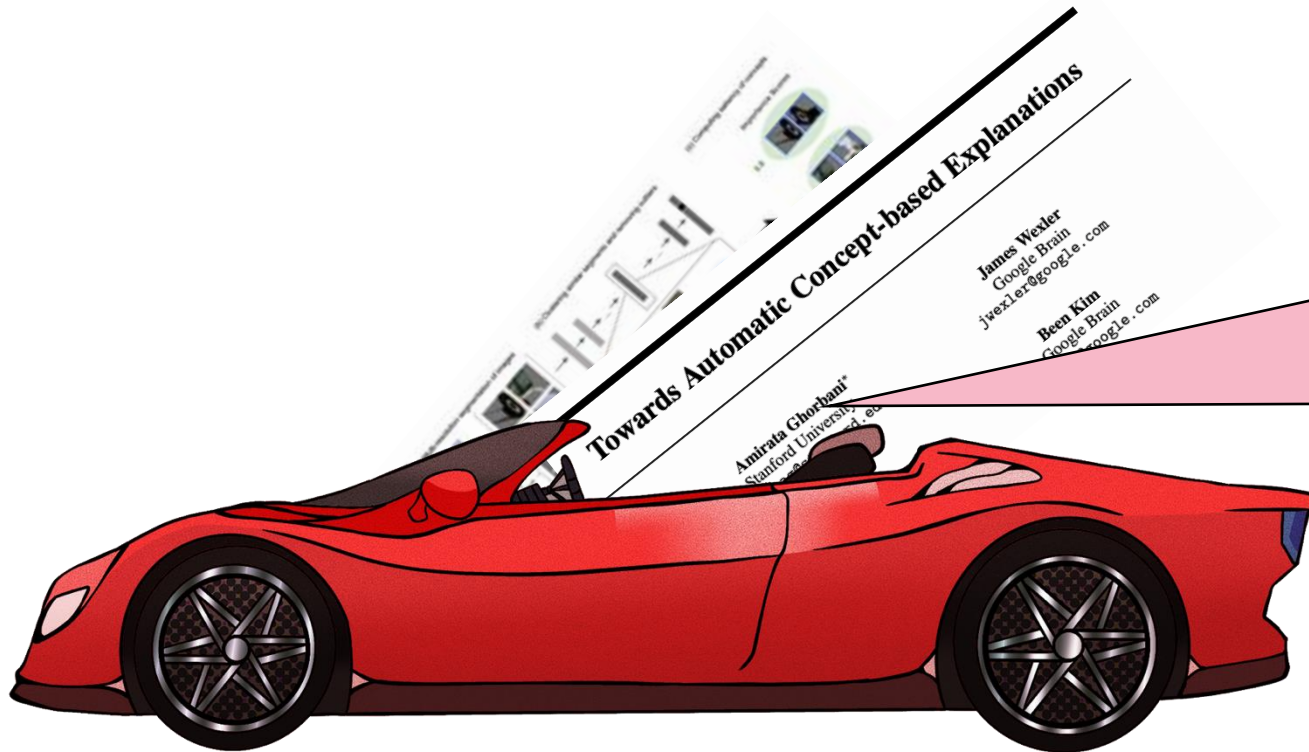


"Wouldn't it be nice if T-CAV concepts could be automatically discovered?"

- Amirata Ghorbani et al. (Probably...)



GOING UNSUPERVISED



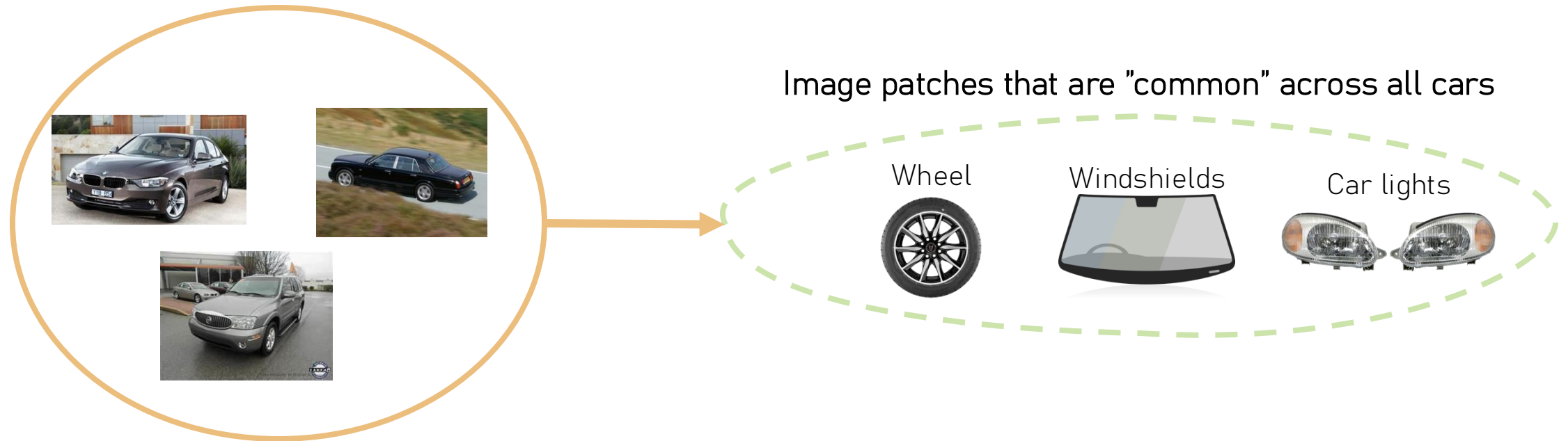
We would like to discover concepts that are:

1. Meaningful
2. Coherent
3. Important

TOWARDS AUTOMATIC CONCEPT EXTRACTION

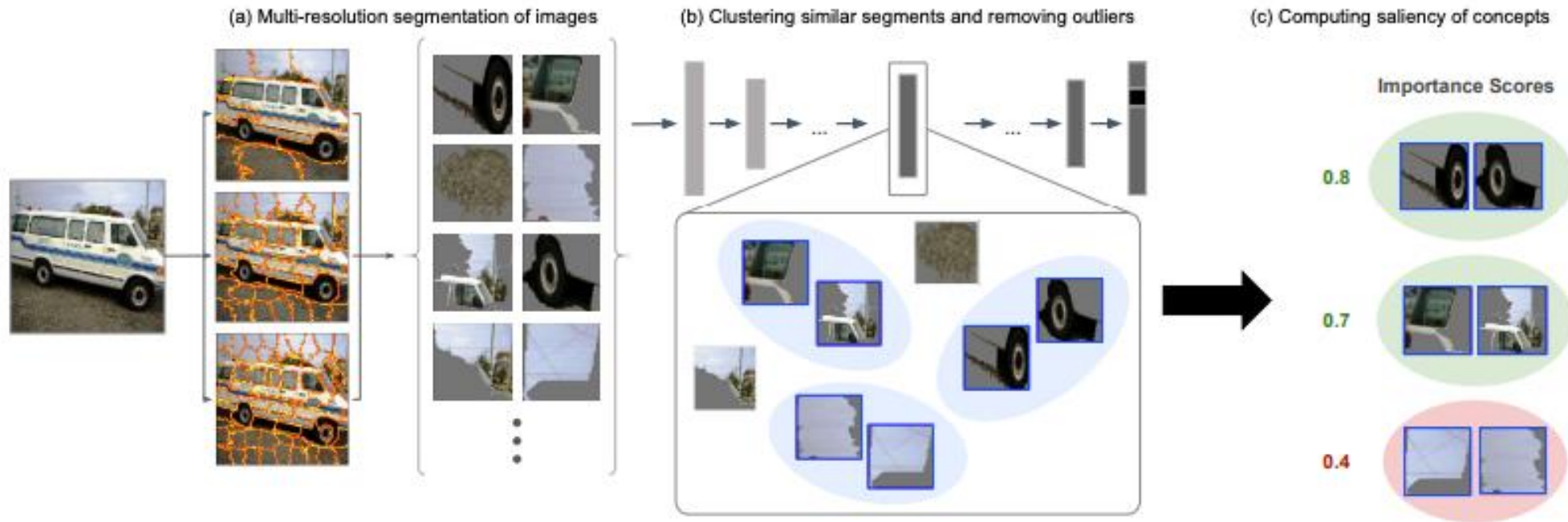
Main Idea: patches of pixels found across images can be thought of as concepts!

Training examples for class "car"



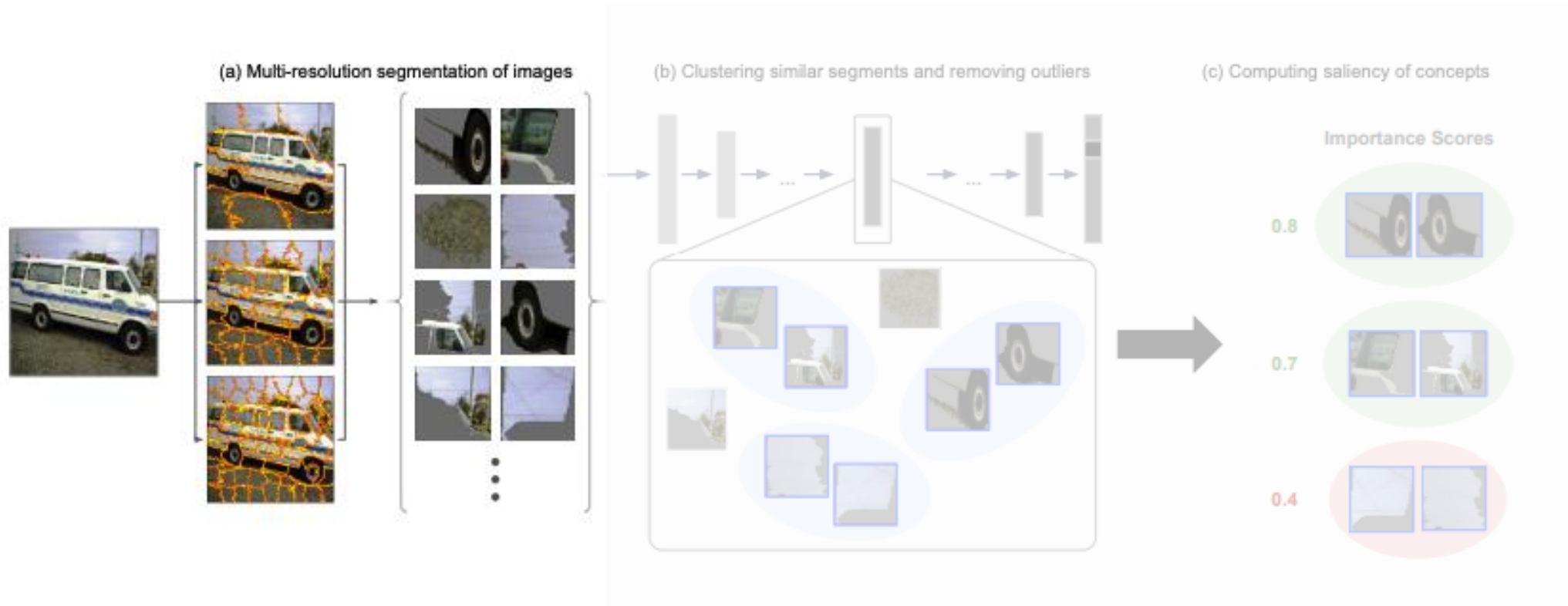
TOWARDS AUTOMATIC CONCEPT EXTRACTION

Main Idea: patches of pixels found across images can be thought of as concepts!



Here's an architecture that can do this!

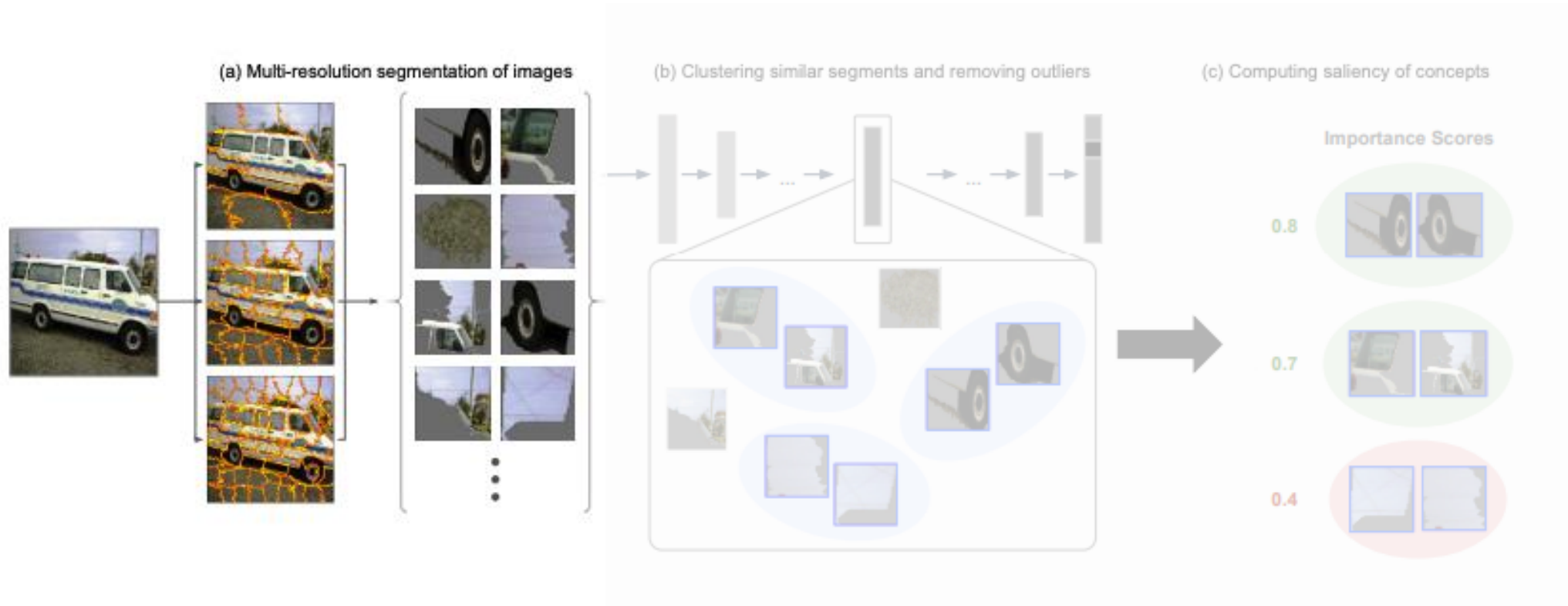
ACE: ARCHITECTURE



Step 1: segment the sample across multiple-resolutions (**why?**)

Desiderata enforced: meaningfulness

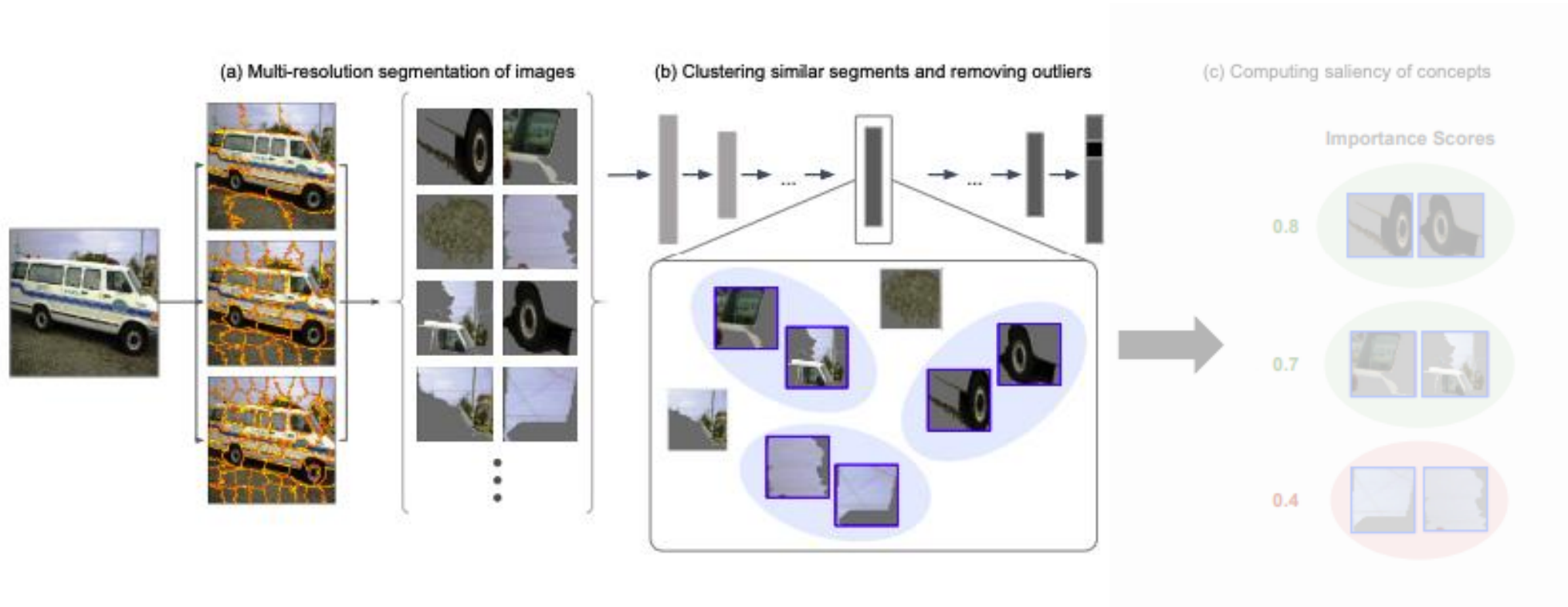
ACE: ARCHITECTURE



Step 1: segment the sample across multiple-resolutions (**why?**)
ACE uses SLIC [1] , a fast* multi-resolution segmentation algorithm.

Desiderata enforced: meaningfulness

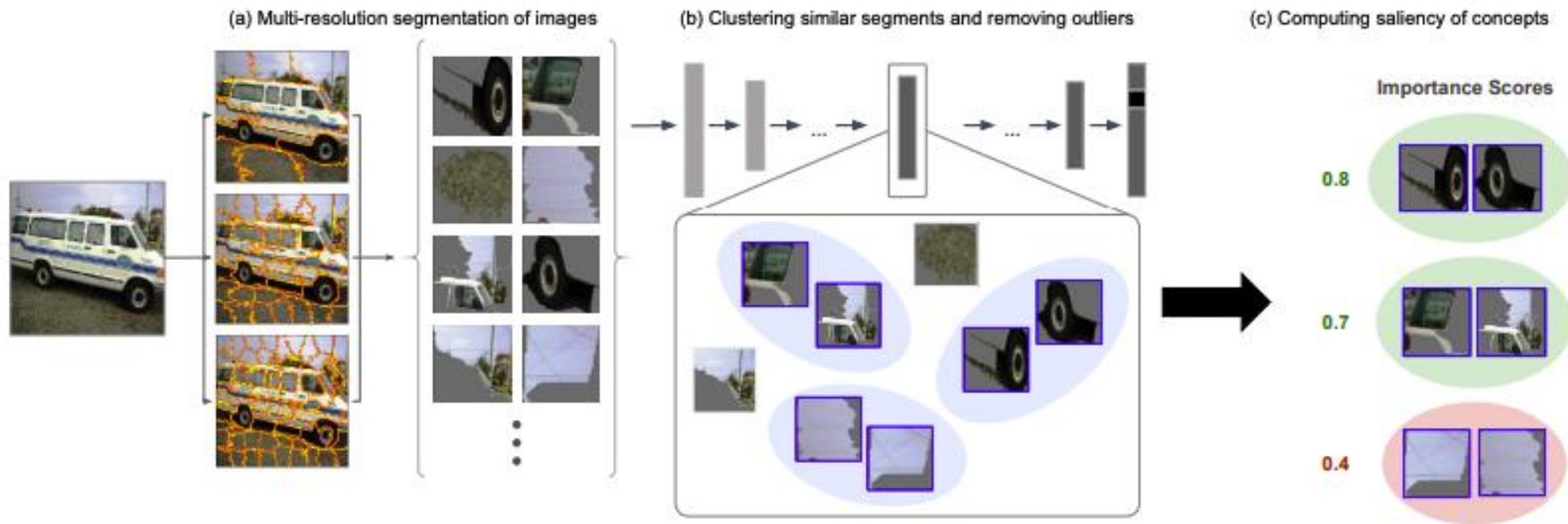
ACE: ARCHITECTURE



Step 2: cluster extracted segments using a hidden layer of a CNN as a feature extractor (**why?**). Then get rid of outliers as these are not useful concepts.

Desiderata enforced: coherence

ACE: ARCHITECTURE



Step 3: use T-CAV with the newly discovered concepts to explain the prediction of the sample of interest!

Desiderata enforced: importance

ACE IN THE WILD



What are the most **salient discovered concepts** for some of the ImageNet classes?

ACE IN THE WILD



What are the most **salient discovered concepts** for some of the ImageNet classes?

Why do you think the jersey is a more salient concept for “Basketball” than the actual ball?

ACE IN THE WILD



What are the most **salient discovered concepts** for some of the ImageNet classes?

ACE has also been generalised to learn concepts in Graph Neural Networks in GCExplainer (Magister et al. 2021) [2]



[1] Figure adapted from: Ghorbani, Amirata, et al. "Towards automatic concept-based explanations." *Advances in Neural Information Processing Systems* 32 (2019).

[2] Magister, Lucie Charlotte, et al. "GCExplainer: Human-in-the-Loop Concept-based Explanations for Graph Neural Networks." *arXiv preprint arXiv:2107.11889* (2021).

GROUNDING ACE

ACE's hyperparameters and processing steps have **several limitations**:

1. We can never be certain that we properly **cover all useful concepts**



Important concepts for **underrepresented populations** could be removed as outliers!

GROUNDING ACE

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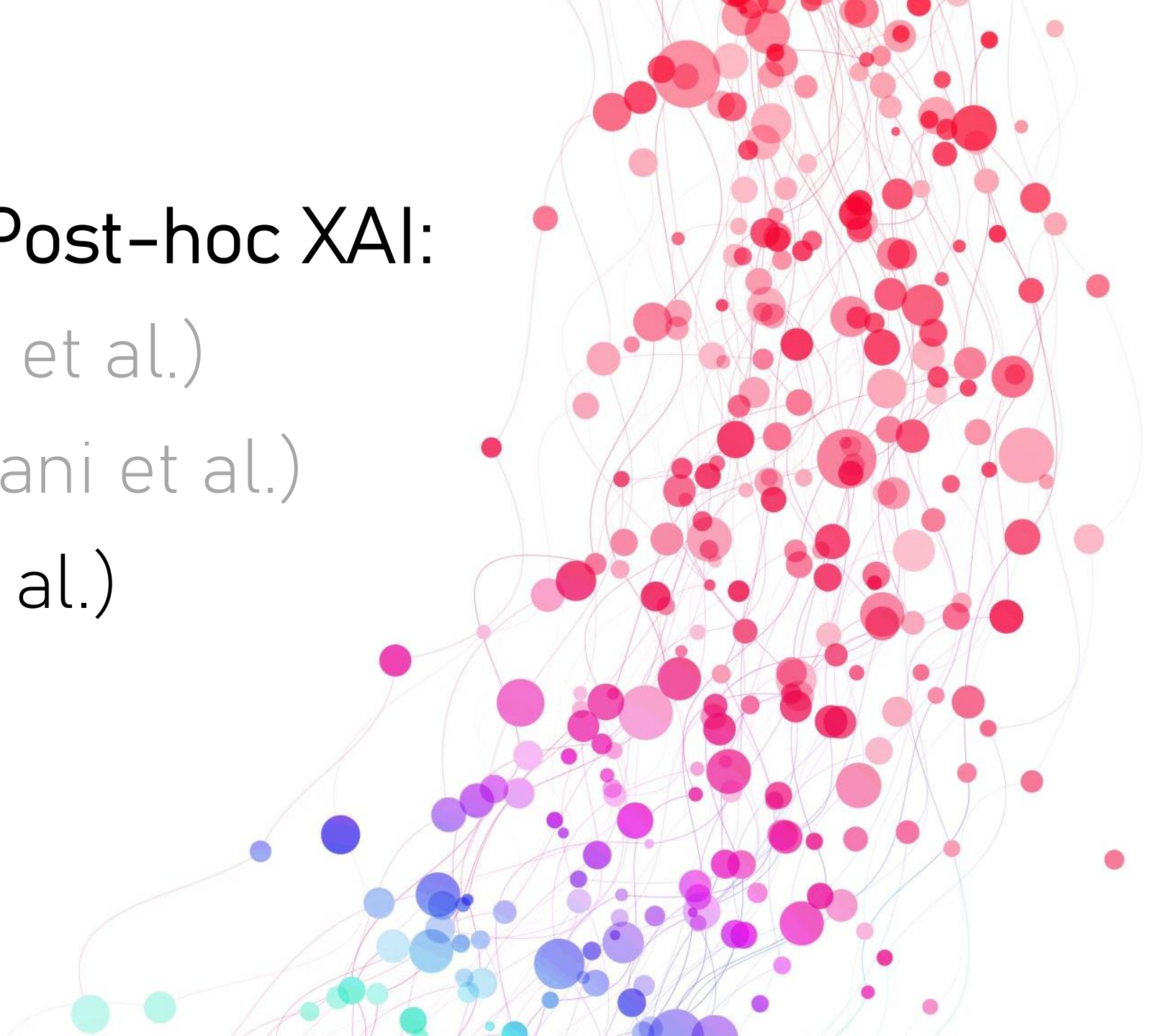
1. We can never be certain that we properly **cover all useful concepts**
2. We won't detect concepts that **interact non-linearly** with the output labels

Looking at the gradients provides understanding of **local (linear) sensitivity**

$$S_{C,k,l}(x) = \nabla h_{l,k}(f_l(x)) \cdot v_C^l$$

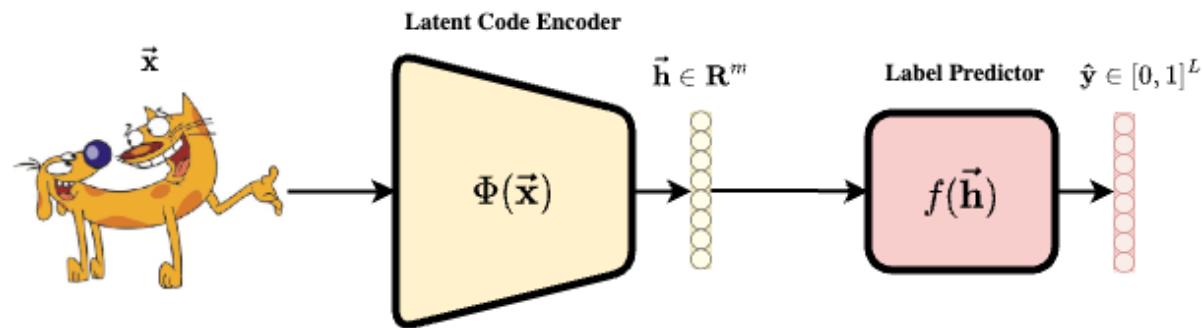
Concept-based Post-hoc XAI:

- T-CAV (Kim et al.)
- ACE (Ghorbani et al.)
- CCE (Yeh et al.)



COMPLETENESS-AWARE CONCEPT EXTRACTION

CCE [1] explains a DNN $\psi(\mathbf{x})$ by discovering a **complete set of concepts**.



We assume that $\psi(\mathbf{x})$ can be decomposed into:

1. A mapping Φ from the inputs $\mathbf{x}$ to an intermediate hidden layer $\Phi(\mathbf{x})$; and
2. A mapping f from that intermediate hidden layer $\Phi(\mathbf{x})$ to the output layer's prediction.



LEARNING CONCEPT MATRICES

Main Idea: learn a **matrix of concept vectors** $\mathbf{C} \in \mathbb{R}^{k \times m}$ and use a “*concept completeness score*” to measure their completeness:

$$n_f(\mathbf{c}_1, \dots, \mathbf{c}_m) = \frac{\sup_g \mathbb{P}_{x,y \sim V} \left[y = \operatorname{argmax}_{y'} f_{y'} \left(g(\mathbf{C} \phi(x)) \right) \right] - a_r}{\mathbb{P}_{x,y \sim V} \left[y = \operatorname{argmax}_{y'} f_{y'}(\mathbf{x}) \right] - a_r}$$

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Then, update $\mathbf{C}$ by optimising this score!

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Looks very scary but it is saying a very simple thing:

“If I project the hidden state into the concept space defined by $\mathbf{C}$, can I faithfully reconstruct it afterwards?”

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Then, update $\mathbf{C}$ by optimising this score!

It measures “faithfulness” by looking at the DNN’s accuracy when the hidden layer is replaced by its reconstruction!

COMPLETENESS VIA RECONSTRUCTION

We define a concept set's "completeness score" w.r.t. DNN $\psi(\mathbf{x})$ as:

$$\text{ConceptCompleteness}(\mathbf{c}_1, \dots, \mathbf{c}_m) := \frac{\sup_{(x,y) \sim \mathcal{D}} \text{Acc}_{(x,y) \sim \mathcal{D}} \left(f \left(g(\mathbf{C} \Phi(\mathbf{x})) \right), y \right) - \text{Acc}_{(x,y) \sim \mathcal{D}}(\text{random label}, y)}{\text{Acc}_{(x,y) \sim \mathcal{D}}(f(\Phi(\mathbf{x})), y) - \text{Acc}_{(x,y) \sim \mathcal{D}}(\text{random label}, y)}$$

Where:

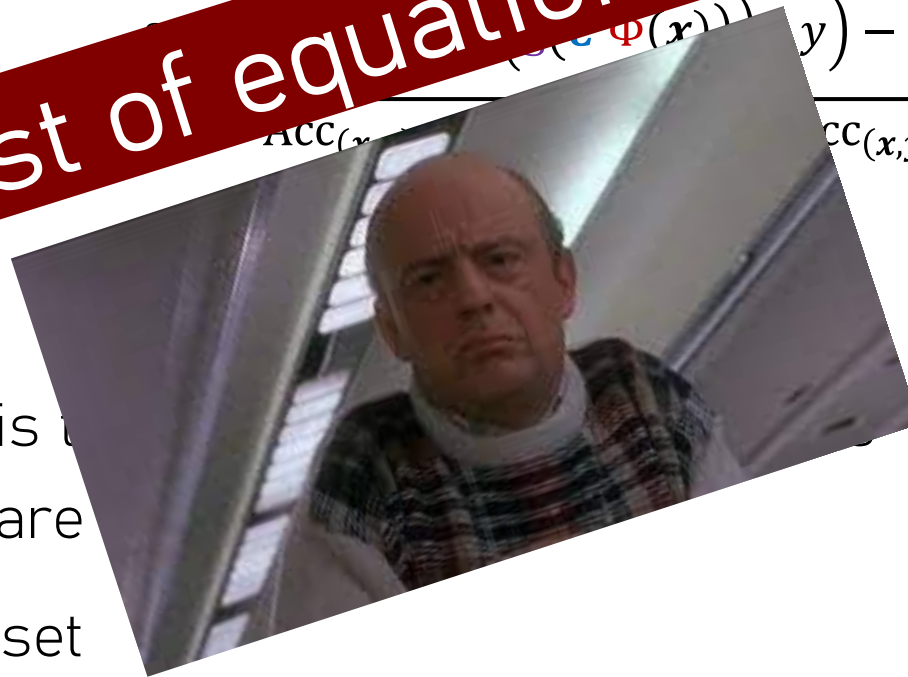
1. $\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m]^T$ is the matrix of concept vectors $\rightarrow$ this is the set of concept vectors we are evaluating!
2. $\mathcal{D}$ is our testing dataset

COMPLETENESS VIA RECONSTRUCTION

We define a concept set's "completeness score" wrt $\mathcal{D}$

$$\text{ConceptCompleteness}(\mathbf{C}) = \frac{\text{Acc}_{(x,y) \sim \mathcal{D}}(\phi(\mathbf{C}, \mathbf{x})) - \text{Acc}_{(x,y) \sim \mathcal{D}}(\text{random label}, y)}{\text{Acc}_{(x,y) \sim \mathcal{D}}(\text{random label}, y)}$$

- W
1. $\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m]^T$ is the set of concept vectors we are considering $\rightarrow$ this is the set of
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We define a concept set's "completeness score" w.r.t. DNN $\psi(\mathbf{x})$ as:

Can we learn a model g that can reconstruct the hidden layer from the concept scores?

$$\text{ConceptCompleteness}(\mathbf{c}_1, \dots, \mathbf{c}_m) := \frac{\sup_{(x,y) \sim \mathcal{D}} \left(\overbrace{f\left(g(\mathbf{c} \Phi(x))\right)}^{\text{reconstruction}} \right), y) - \text{Acc}_{(x,y) \sim \mathcal{D}}(\text{random label}, y)}{\text{Acc}_{(x,y) \sim \mathcal{D}}(f(\Phi(x)), y) - \text{Acc}_{(x,y) \sim \mathcal{D}}(\text{random label}, y)}$$

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And adjust for the accuracy of a random (why?)

COMPLETENESS VIA RECONSTRUCTION

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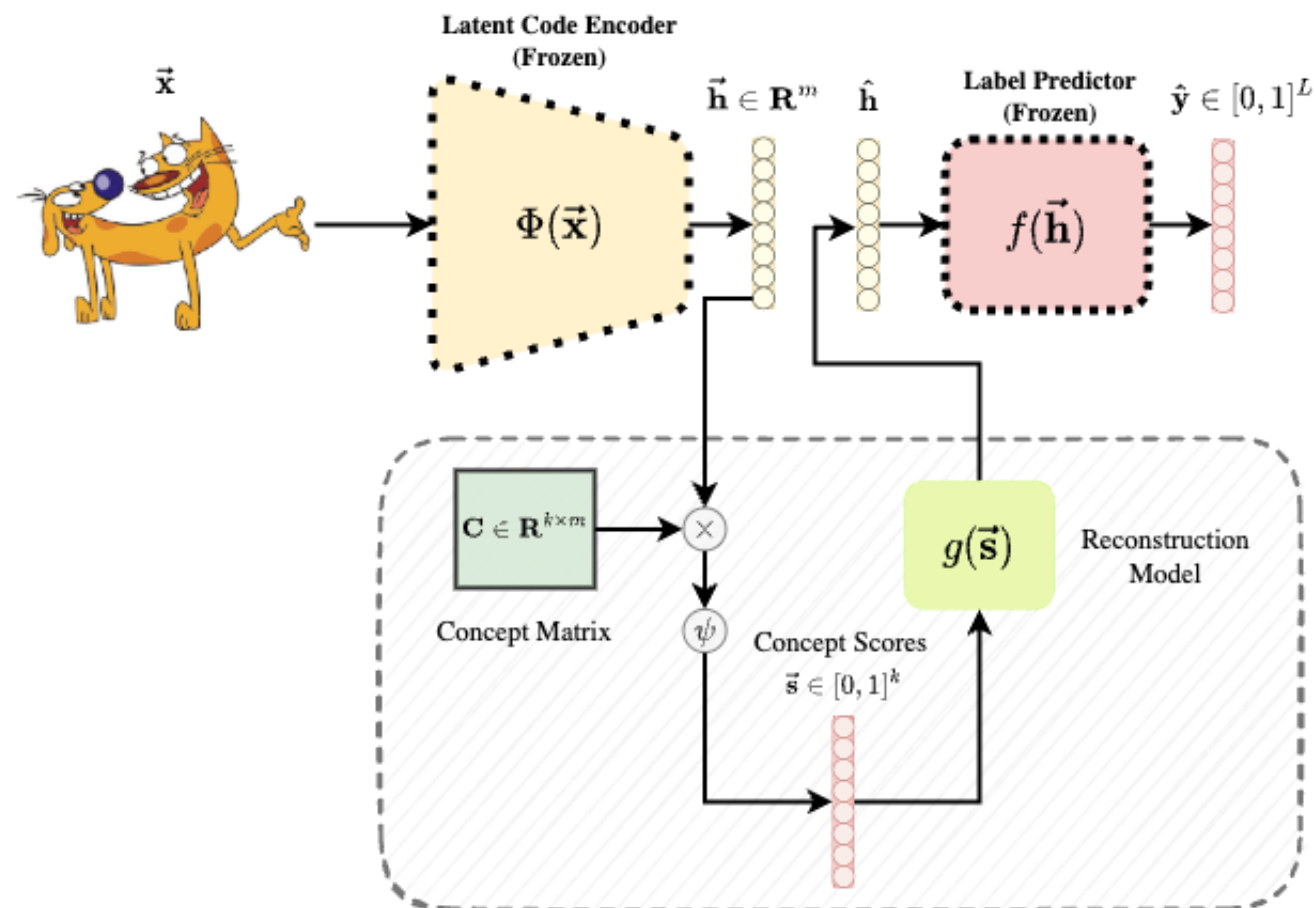
Let's learn a set of concept vectors that maximises this metric!

COMPLETENESS-AWARE CONCEPT EXTRACTION

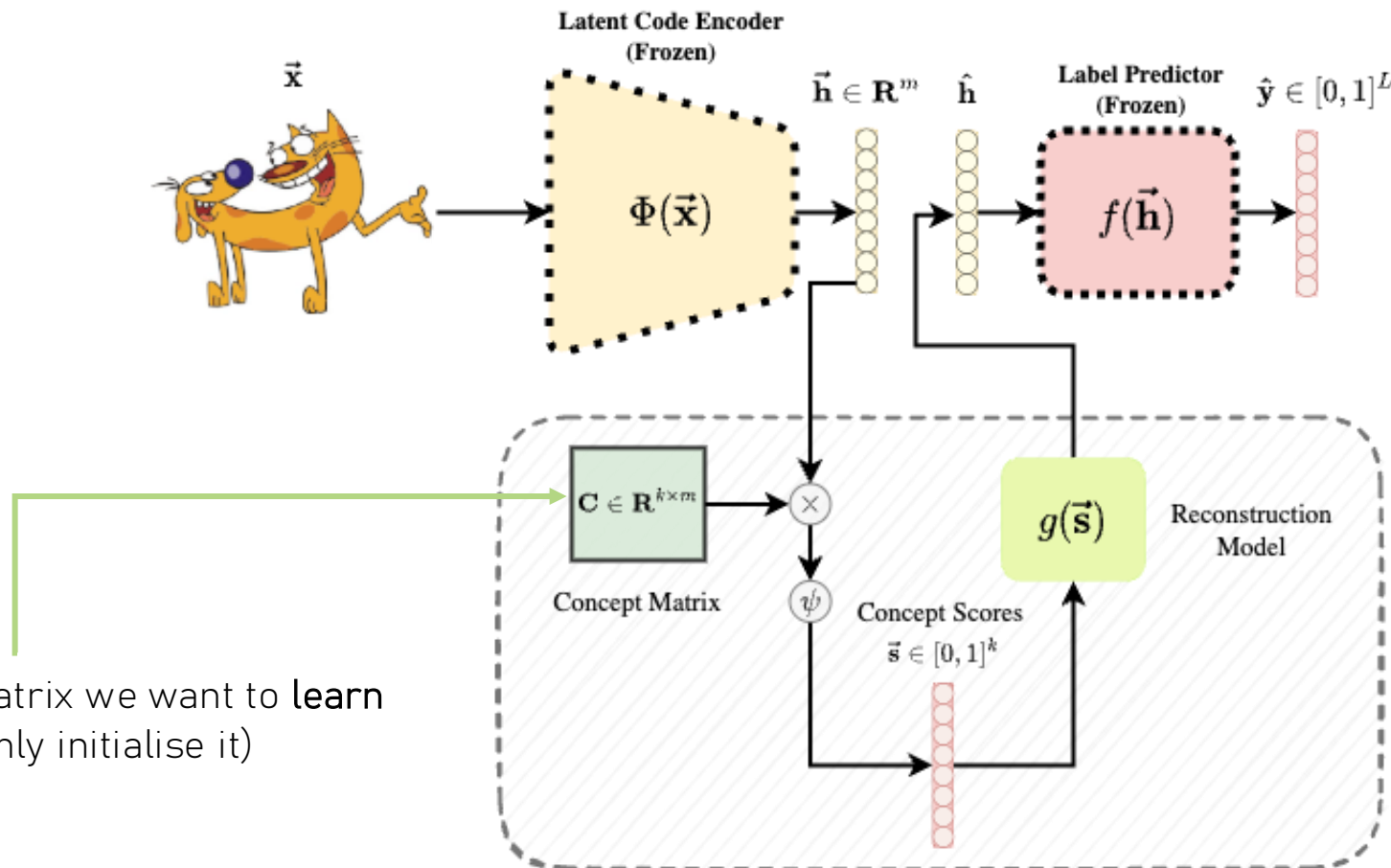
We want to learn k concept vectors $\mathcal{C} \in \mathbb{R}^{k \times m}$ such that:

1. Each vector represents a **distinct concept direction**
2. When a hidden layer of the input DNN is projected into the concept space, their resulting score **preserves all the information needed** to reconstruct the hidden layer.

COMPLETENESS-AWARE CONCEPT EXTRACTION

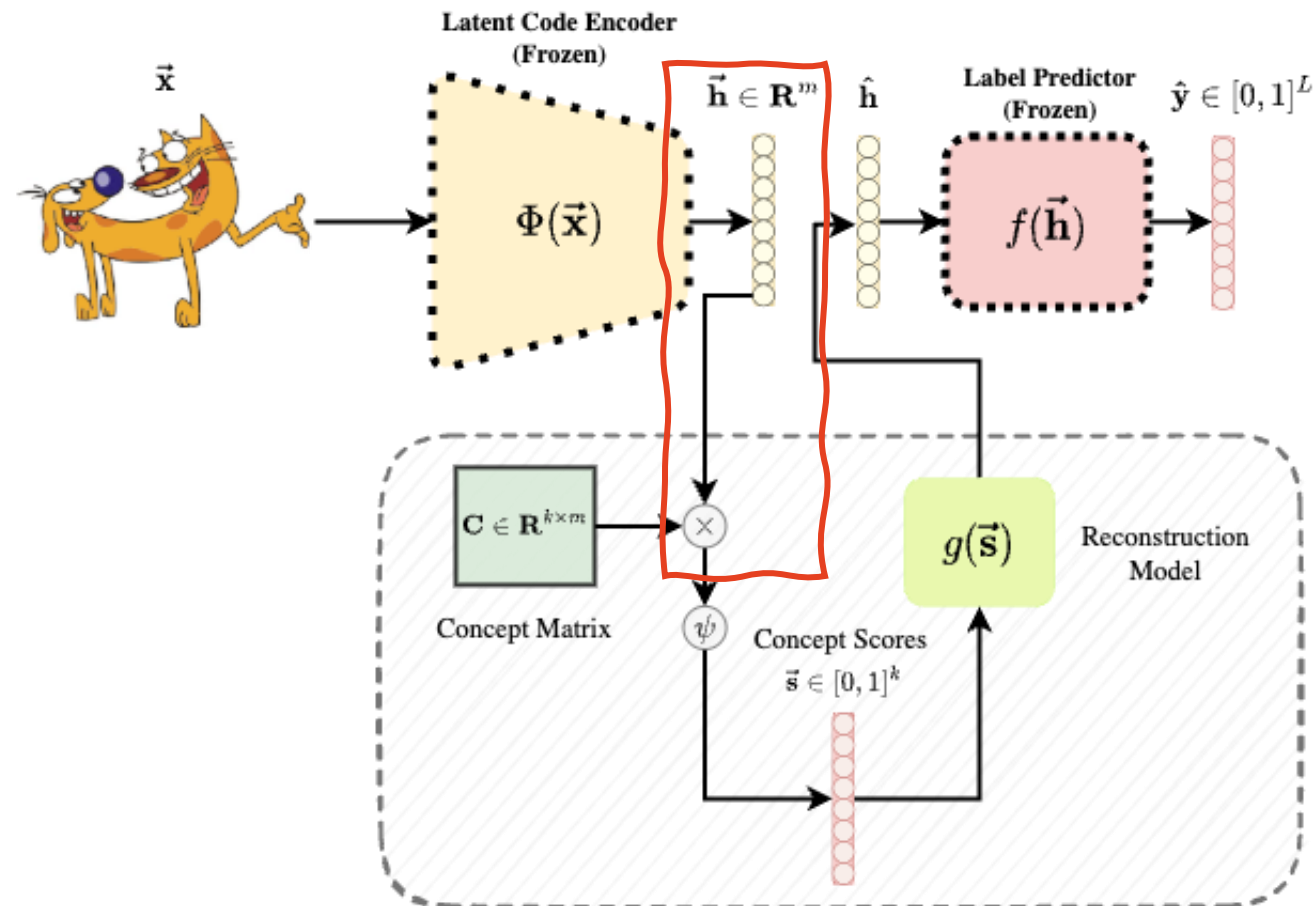


COMPLETENESS-AWARE CONCEPT EXTRACTION



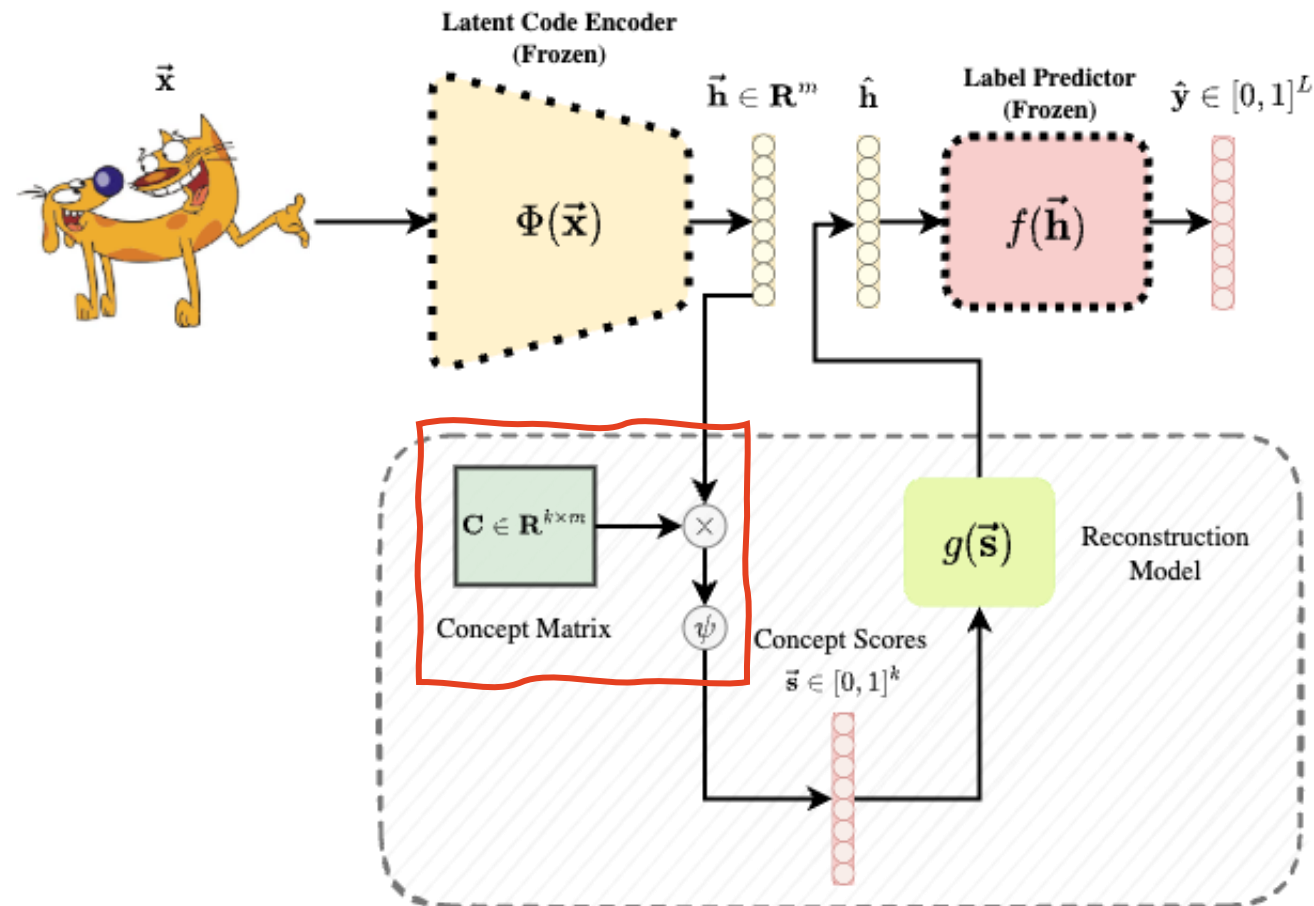
This is the concept matrix we want to learn
(originally, we randomly initialise it)

COMPLETENESS-AWARE CONCEPT EXTRACTION



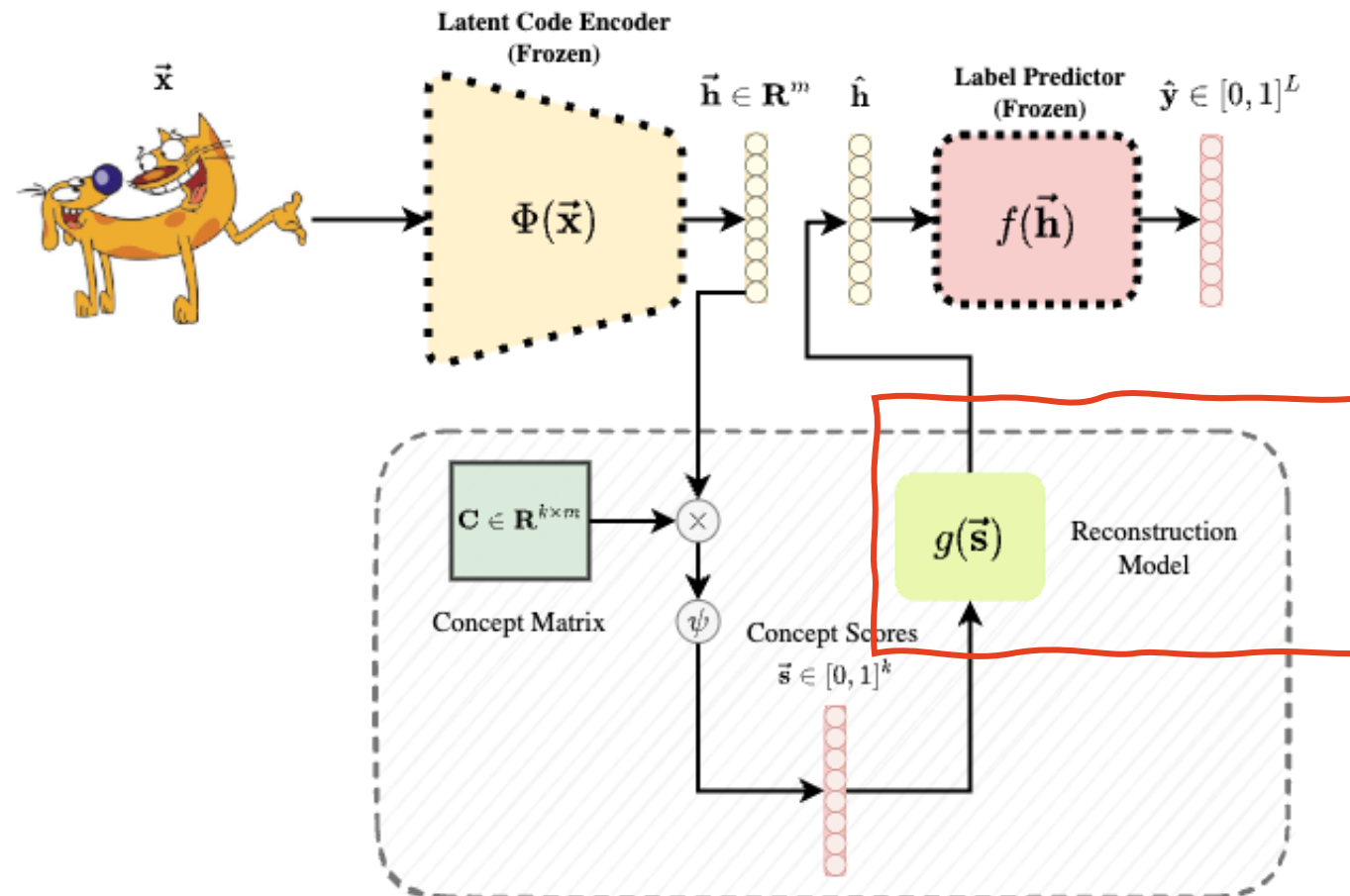
Step 1: project the DNN's hidden layer into the concept space

COMPLETENESS-AWARE CONCEPT EXTRACTION



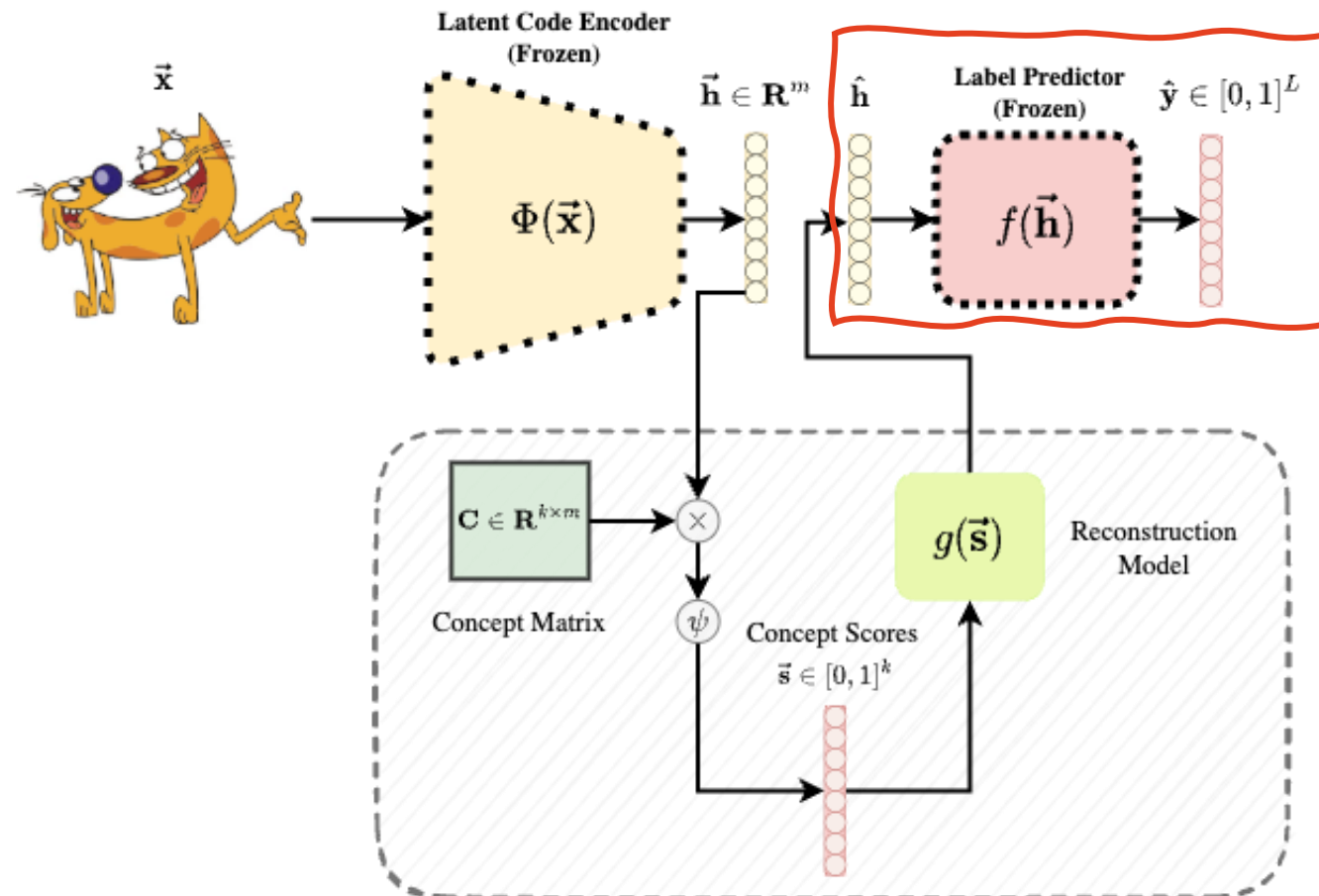
Step 2: compute a set of concept scores by thresholding and normalising the concept projection

COMPLETENESS-AWARE CONCEPT EXTRACTION



Step 3: pass the concepts scores to a learnable model $g(\vec{s}) = \hat{h}$ that aims to reconstruct $\vec{h}$ from $\vec{s}$

COMPLETENESS-AWARE CONCEPT EXTRACTION



Step 4: use $\hat{h}$ as the reconstructed hidden layer and predict an output class using the rest of the DNN

CONCEPT DIVERSITY AND COHERENCE

CCE further encourages discovered concepts to be:

1. **Coherent**: similar samples should remain close in concept-space

$$R(\mathbf{c}) = \lambda_1 \frac{\sum_{k=1}^m \sum_{\mathbf{x}_a^b \in T_{\mathbf{c}_k}} \Phi(\mathbf{x}_a^b) \cdot \mathbf{c}_k}{mK} - \lambda_2 \frac{\sum_{j \neq k} \mathbf{c}_j \cdot \mathbf{c}_k}{m(m-1)}$$

CONCEPT DIVERSITY AND COHERENCE

CCE further encourages discovered concepts to be:

1. **Coherent**: similar samples should remain close in concept-space
2. **Diverse**: concept vectors should be as distinct from each other as possible

$$R(\mathbf{c}) = \lambda_1 \frac{\sum_{k=1}^m \sum_{\mathbf{x}_a^b \in T_{\mathbf{c}_k}} \Phi(\mathbf{x}_a^b) \cdot \mathbf{c}_k}{mK} - \lambda_2 \frac{\sum_{j \neq k} \mathbf{c}_j \cdot \mathbf{c}_k}{m(m-1)}$$

CONCEPT CONTRIBUTIONS

We can assign each concept a **score** that fairly represents how much they contributed towards the completeness score

$$\mathbf{s}_i(\eta) = \sum_{S \subseteq C_s \setminus \mathbf{c}_i} \frac{(m - |S| - 1)!|S|!}{m!} [\eta(S \cup \{\mathbf{c}_i\}) - \eta(S)]$$

Does this look familiar?

CONCEPT CONTRIBUTIONS

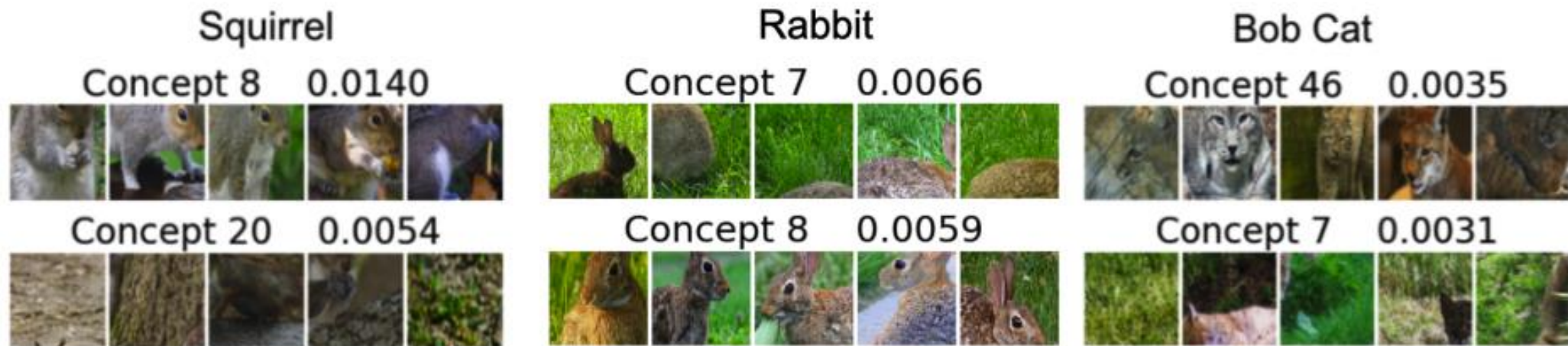
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These are Shapley Scores (i.e., **ConceptSHAP**)

CCE: APPLICATIONS

In vision, CCE can discover meaningful human-understandable concepts:



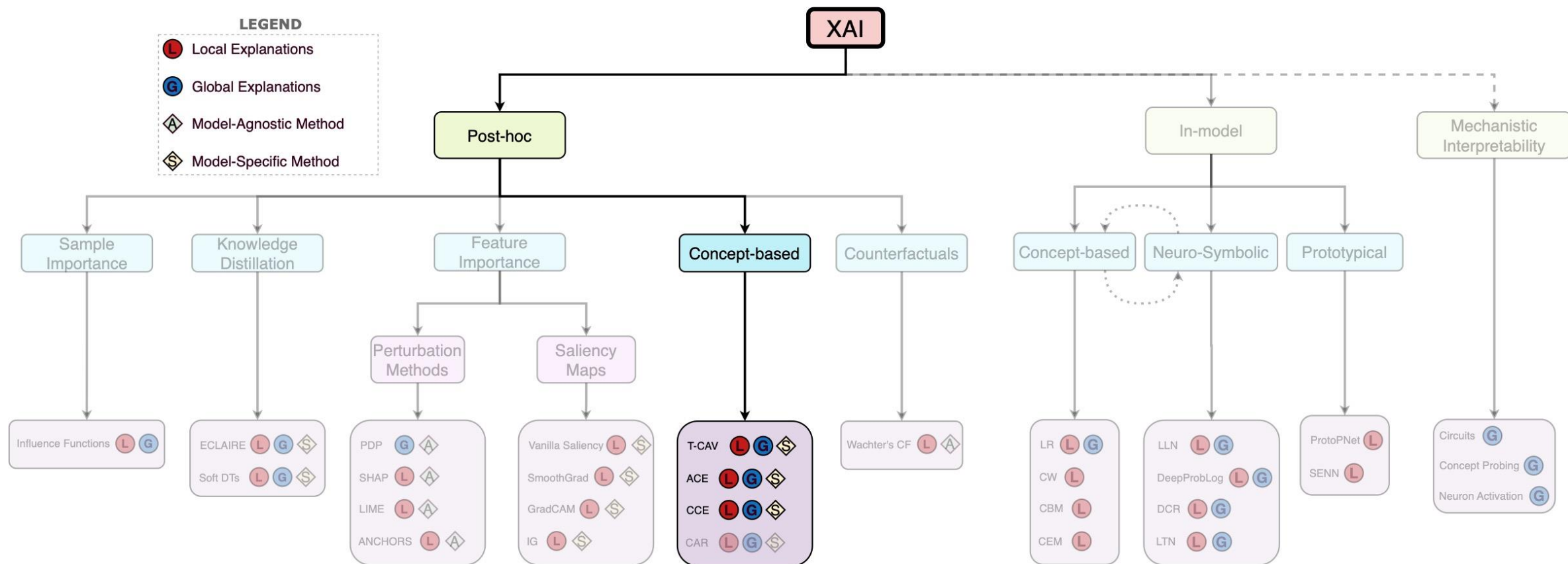
CCE: APPLICATIONS

And it can be applied to other data modalities like text!

Table 2: The 4 discovered concepts and some nearest neighbors along with the most frequent words that appear in top-500 nearest neighbors.

Concept	Nearest Neighbors	Frequent words	ConceptSHAP
1	poorly constructed what comes across as interesting is the wasting my time with a comment but this movie awful in my opinion there were <UNK> and the	worst (168) ever (69) movie (61) seen (55) film (50) awful (42) time(40) waste (34) poorly (26) movies (24) films (18) long (17)	0.280
2	normally it would earn at least 2 or 3 <UNK> <UNK> is just too dumb to be called i feel like i was ripped off and hollywood	not (58) movie (39) make (25) too (23) film (22) even (19) like (18) 2 (16) never (14) minutes (13) 1 (12) doesn't (11)	0.306
3	remember awaiting return of the jedi with almost <UNK> better than most sequels for tv movies i hate male because marie has a crush on her attractive	movies (19) like (18) see (16) movie (15) love (15) good (12) character (11) life (11) little (10) ever (9) watch (9) first (9)	0.174
4	new <UNK> <UNK> via <UNK> <UNK> with absolutely hilarious homosexual and an italian clown <UNK> is an entertaining stephen <UNK> on the vampire <UNK> as a masterpiece	excellent (50) film (25) perfectly (19) wonderful (19) perfect (16) hilarious (15) best (13) fun (12) highly (11) movie (11) brilliant (9) old (9)	0.141

TODAY IN A NUTSHELL



QUESTIONS?

